

High load RecSys

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Materials by Vadim Kokhtev



MIPT,
spring 2025

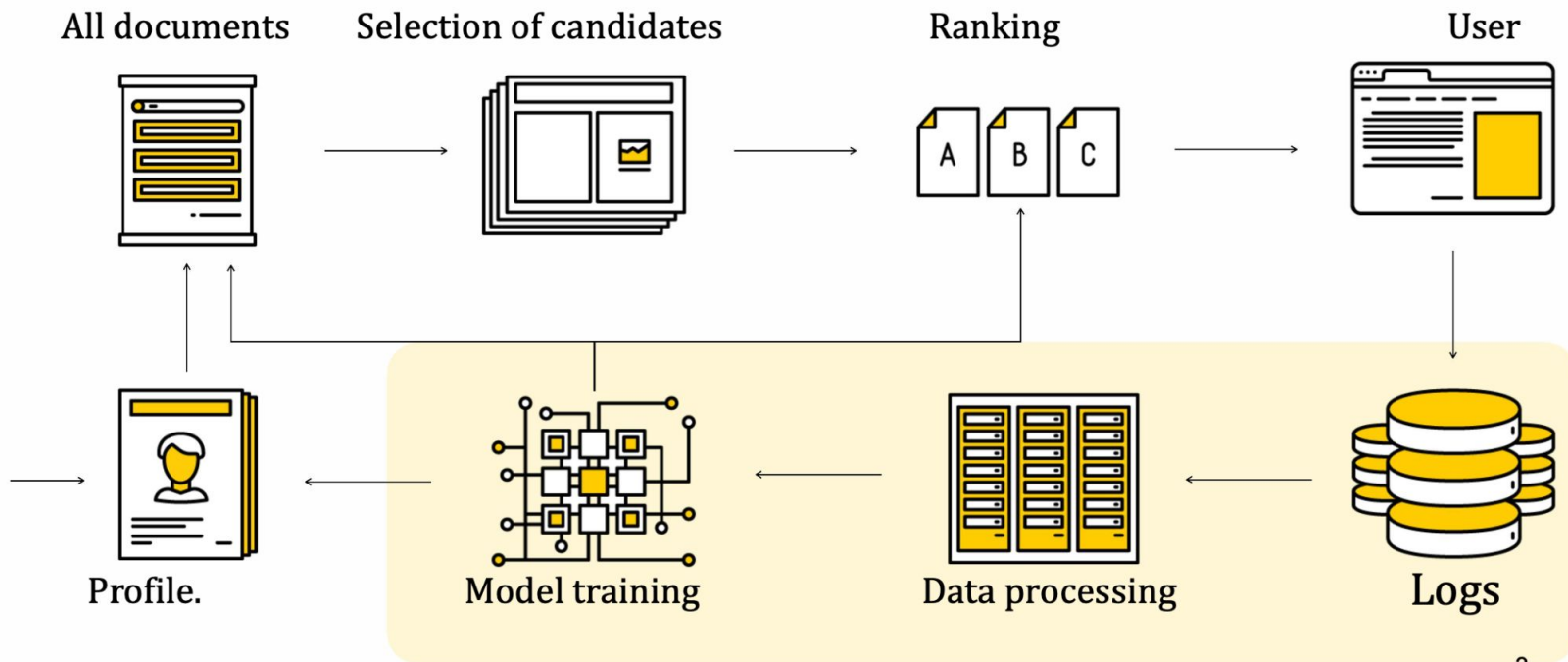
Typical service



- 10^7 documents
- 10^6 users per day
- 1-10k requests per second (RPS) for recommendations

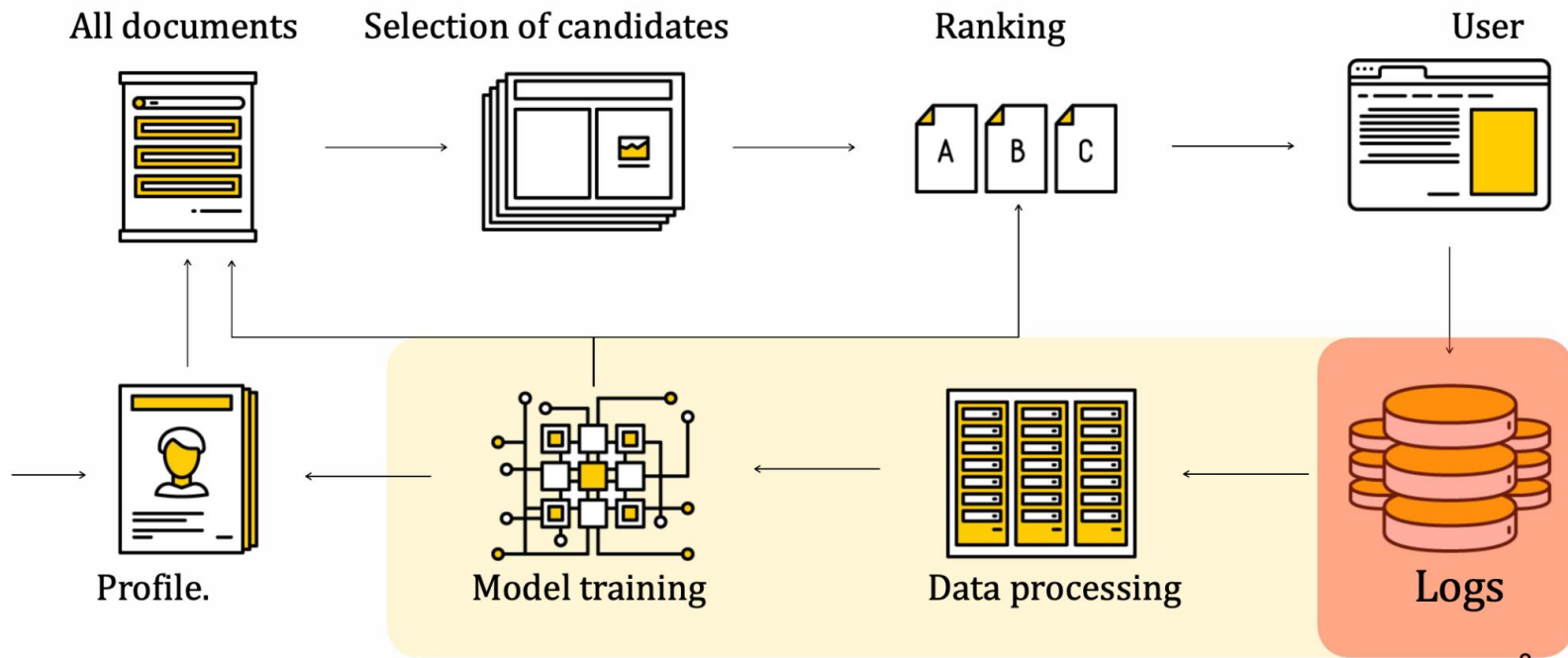
What could possibly go wrong? 🤔

The real system





Logging





Logging

- To train algorithms, information about user actions is stored in a structured form
- At a minimum, you will need a bundle of "user-item-rating-timestamp"
- Logs should be stored forever because this is unique information about your users

<code>_version</code>	<code>String optional</code>
<code>client_info</code>	Struct <i>optional</i> <code>client_type : String</code> <code>client_version : String</code>
<code>event_id</code>	<code>String optional</code>
<code>event_type</code>	<code>String optional</code>

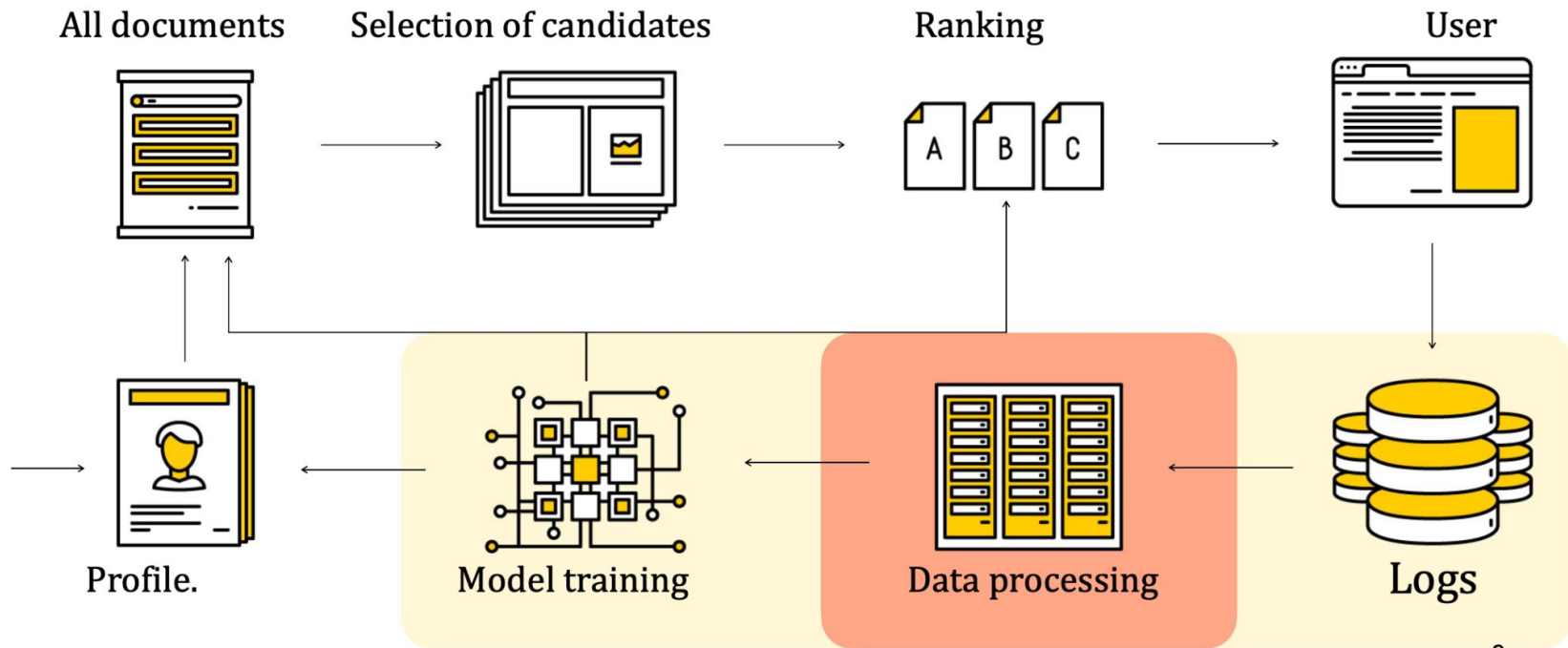
Data processing

girafe
ai

01



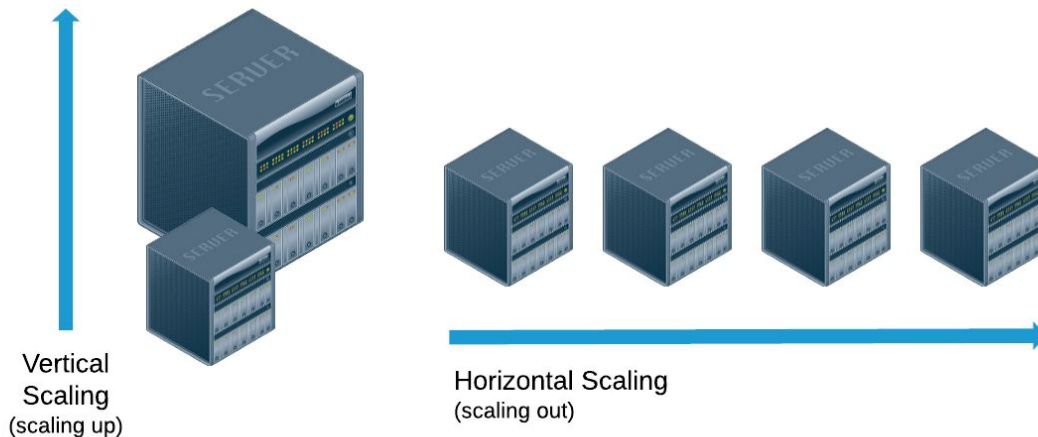
Data processing



Scalability



- When there was less data and they were more structured, they could be put into a relational database, having thought out the data schema and a list of possible queries to them in advance
- To speed up processing, we need a more powerful server - **vertical scalability**
- In the MapReduce paradigm, more average servers are needed to speed up processing (**horizontal scalability**), which is much cheaper





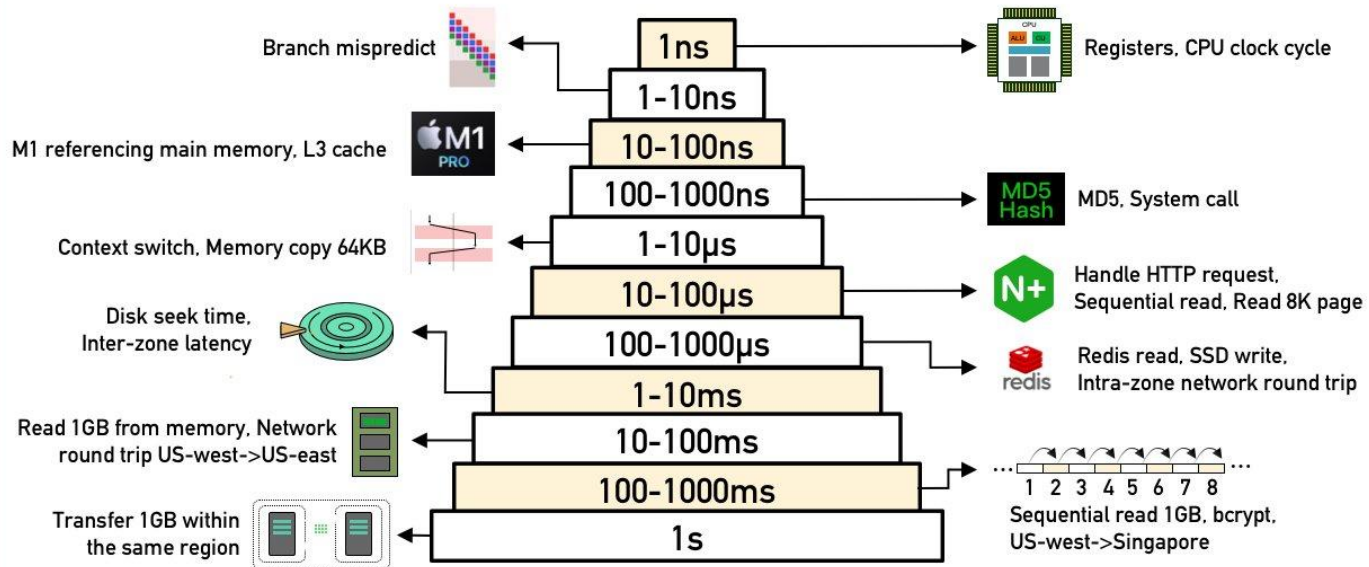
Data Locality

[Ulrich Drepper, What every programmer should know about memory \(2007\)](#)

[Stackoverflow discussion](#)

Latency numbers for the 2020s

ByteByteGo.com





Data Locality

- Blocks of a huge file are stored on different machines
- We can process blocks locally on the machines where they are stored (fast read from local disk)
- If it is possible to process blocks independently in the task, then it is possible to achieve ideal scalability (embarrassingly parallel)
- For example, the task of filtering file lines scales perfectly. All lines are independently checked by the predicate, the more machines, the faster it will work

MapReduce

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Word frequencies counting problem

Texts from all over the Internet are stored in a distributed system.

We are interested to know the frequencies of all words (counters for tf-idf).

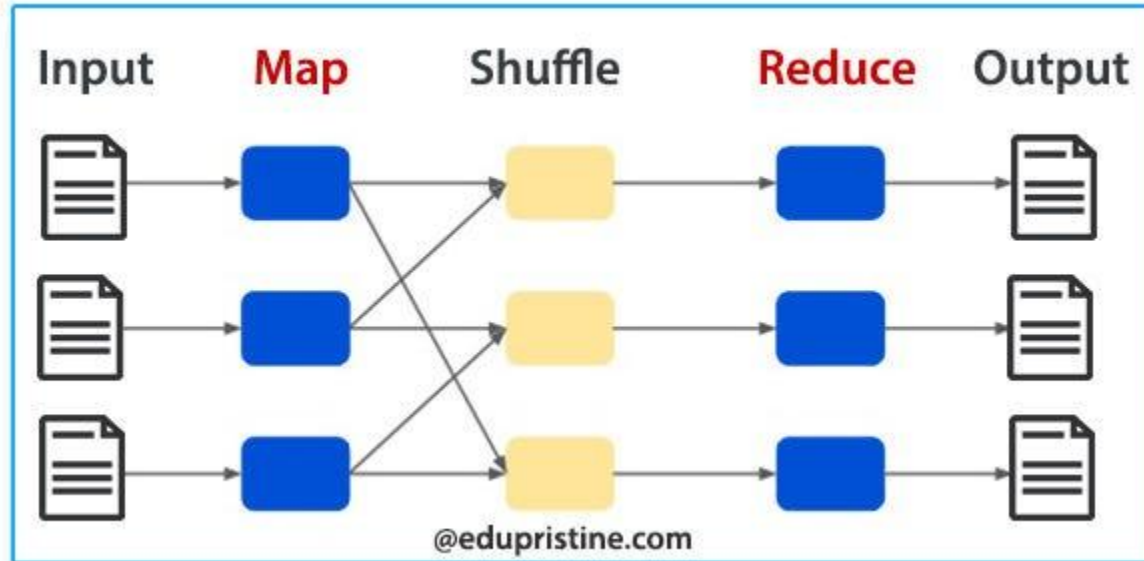
Solution is MapReduce paradigm

How to solve:

- for each block, we calculate the frequency of words in it (ideally scaled)
- add up the frequencies for all blocks



MapReduce main idea





MapReduce solution

Map:

$(K1, V1) \Rightarrow \text{List}(K2, V2)$

(line number, "cat sleeps") $\Rightarrow$ [("cat", 1), ("sleep", 1)]

Shuffle:

Keys are splitted by $\text{hash}(\text{key}) \% N$ on N parts

Each parts gets sorted by **key** (independently)

Reduce:

$(K2, \text{List}(V2)) \Rightarrow \text{List}(K3, V3)$

("cat", (1, 1, 1)) $\Rightarrow$ [("cat", 3)]

Reliability of operations



- If the mapper is lost, we can restart the task only for its blocks
- If the reducer is lost, we re-collect data from all mappers only for its hash value



Popular track example

UserId	TrackId	AlbumId
11123	4521	842
14322	3593	957
...	...	...

Map: $(K1, V1) \Rightarrow \text{List}(K2, V2)$

Reduce: $(K2, \text{List}(V2)) \Rightarrow \text{List}(K3, V3)$

Let's take the logs of listening to music.

How do we find the most popular track for each album?

M: $\#, (\text{user}, \text{track}, \text{album}) \Rightarrow (\text{album}, \text{track}), 1$

R: $(\text{album}, \text{track}), (1, 1, \dots) \Rightarrow (\text{album}, \text{track}), \text{count}$

M: $(\text{album}, \text{track}), \text{count} \Rightarrow \text{album}, (\text{track}, \text{count})$

R: $\text{album}, \text{tracks} \Rightarrow \text{album}, \text{most popular track}$

Example: Item-based CF concurrency

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03

Example: Item-based CF concurrency

- n users, m items, $n \sim 10^6$, $m \sim 10^7$
- It is necessary to calculate in advance the Item-Item similarity

$$s(i, j) = \frac{\sum_{u \in U} (r_{ui} - r_u)(r_{uj} - r_u)}{\sqrt{\sum_{u \in U} (r_{ui} - r_u)^2} \sqrt{\sum_{u \in U} (r_{uj} - r_u)^2}} \quad \begin{array}{l} \text{adjusted} \\ \text{cosine} \\ \text{similarity} \end{array}$$

- Naive approach – $O(m^2n)$
- The evaluation matrix is very sparse, can it be better?
- The contribution to the similarity of the two products is non-zero only from users who rated both of these products



Inverted index

Update the calculations once a day using the inverted index



	2		2	4	5	
	5		4			1
			5		2	
		1		5		4
			4			2
	4	5		1		



$$s(i, j) = \frac{\sum_{u \in U} (r_{ui} - r_u)(r_{uj} - r_u)}{\dots}$$

Map step:

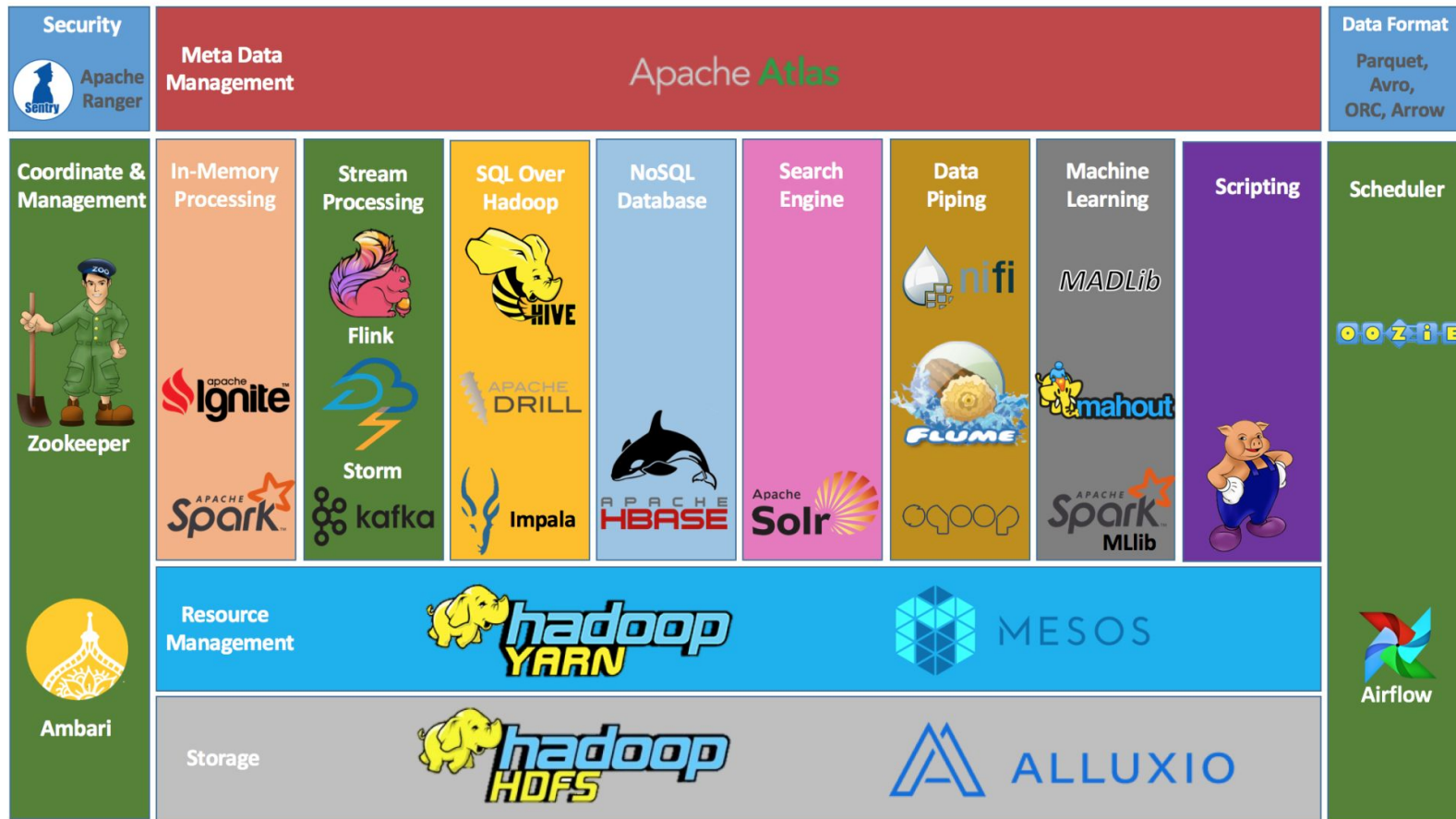
$$(1, 3) \rightarrow (5 - 3.3) * (4 - 3.3)$$

$$(1, 6) \rightarrow (5 - 3.3) * (1 - 3.3)$$

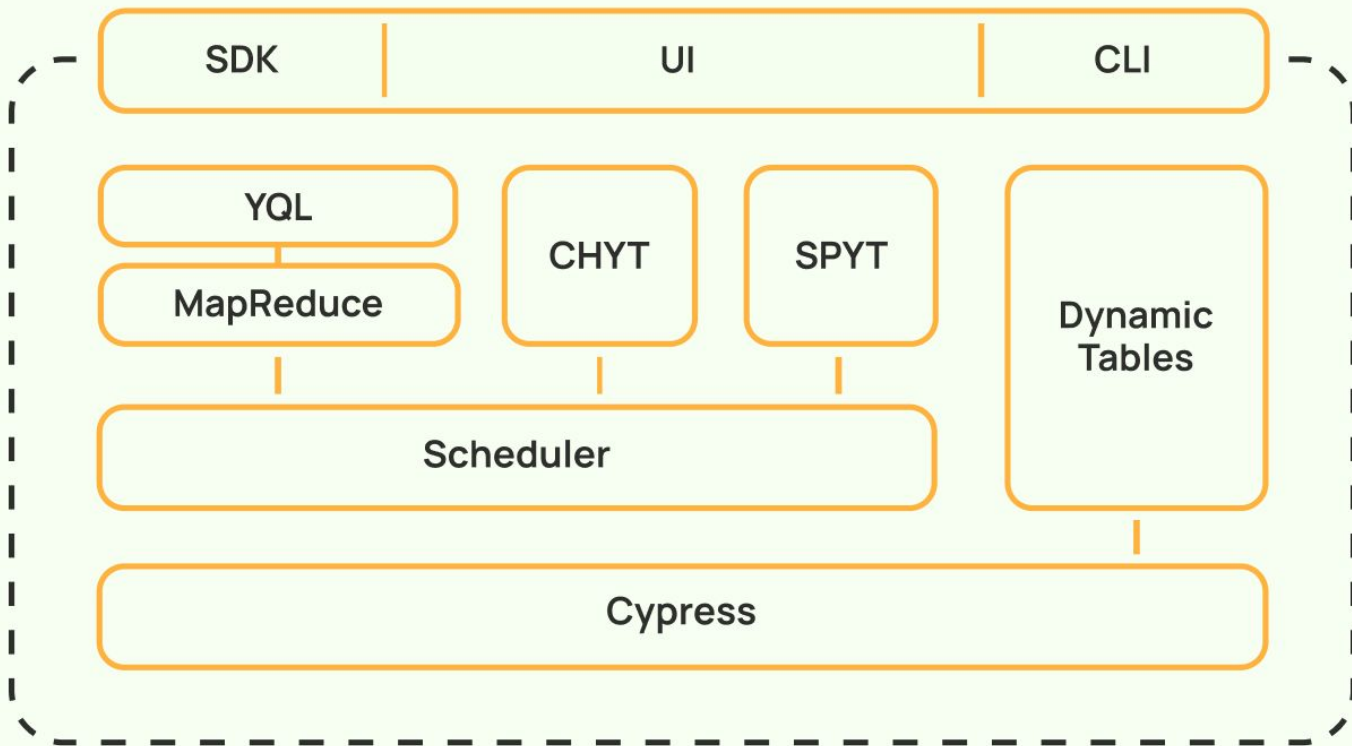
$$(3, 6) \rightarrow (4 - 3.3) * (1 - 3.3)$$

Reduce step: summing up by the key

Hadoop: open source MapReduce

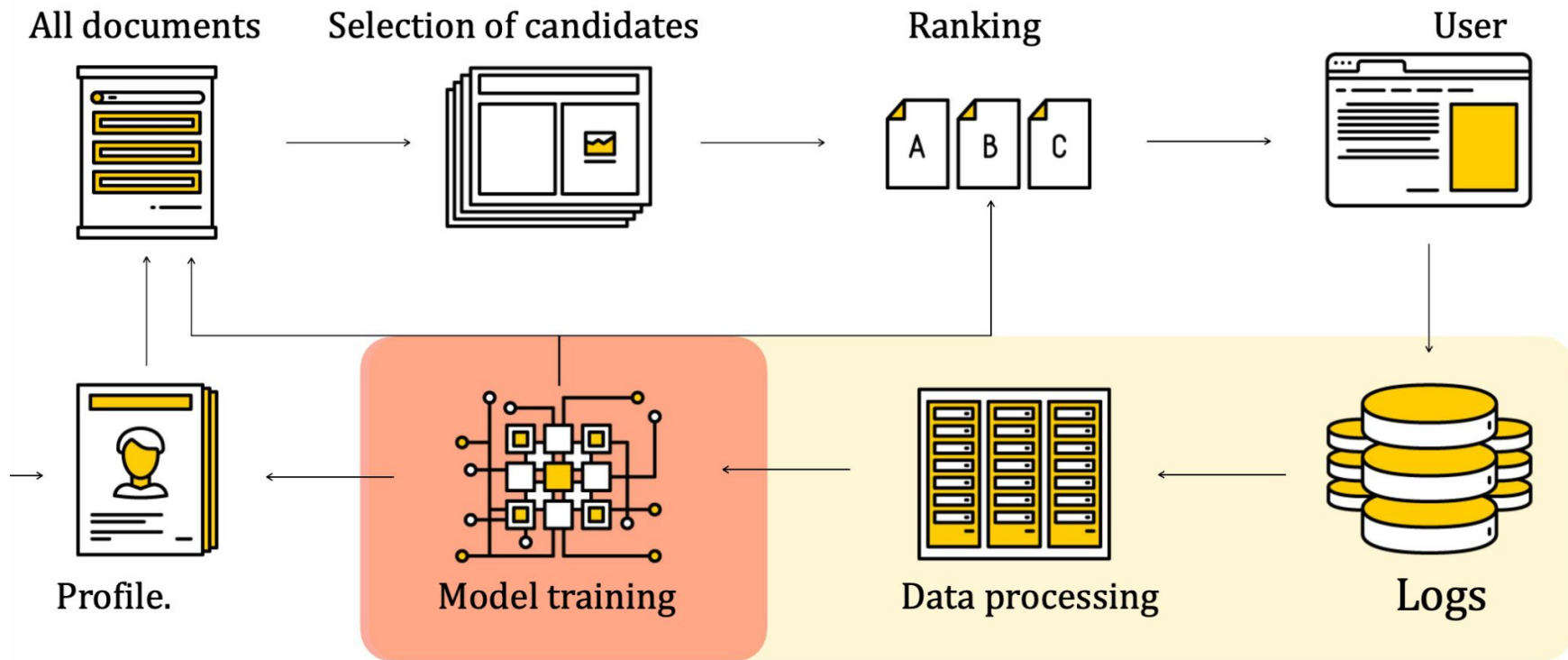


YTsauros: one more MapReduce





Model training



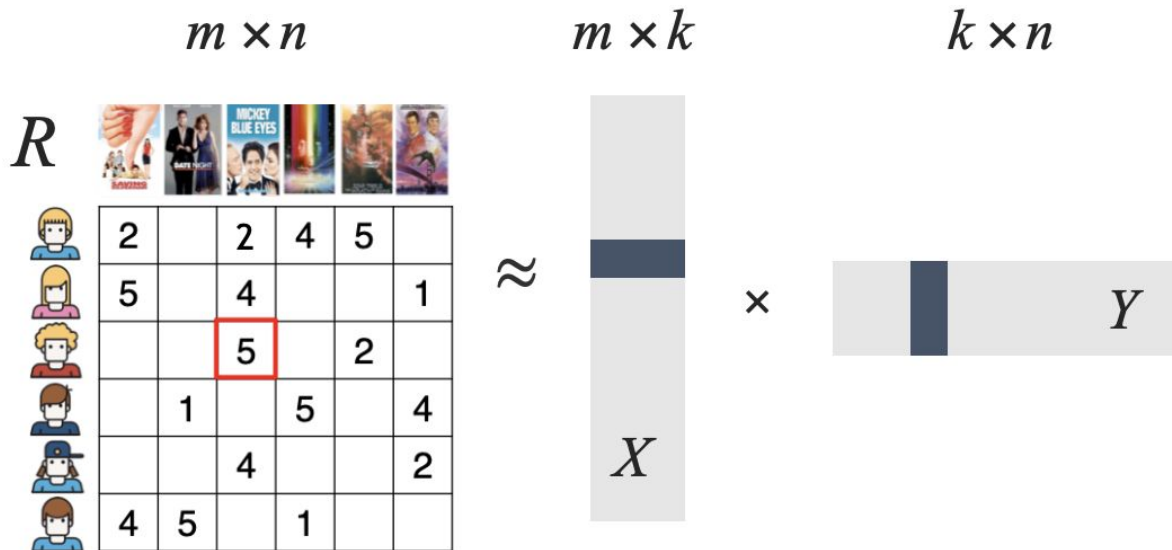
Alternating Least Squares (ALS)

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Alternating Least Squares (ALS)



$$\min_{X, Y} \sum_{(u, i) \in R} (r_{ui} - x_u^T y_i)^2 + \lambda (\|x_u\|^2 + \|y_i\|^2)$$

Alternating Least Squares (ALS)



Initialize X and Y with random values

In the loop:

- We fix the matrix Y (products)
- Find the optimal matrix X (solve the ridge regression for each user)

$$\begin{matrix} \text{User 1 profile} \\ \text{User 2 profile} \\ \text{User 3 profile} \end{matrix} \times \text{Item profile} \approx \begin{matrix} 5 \\ 1 \\ 4 \end{matrix}$$

- and vice versa

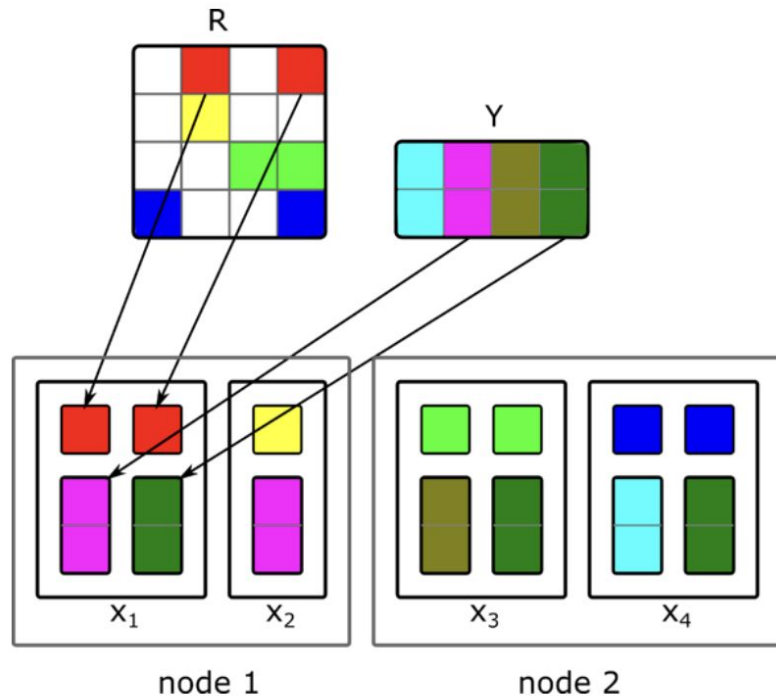
$$\min_{X, Y} \sum_{(u, i) \in R} (r_{ui} - x_u^T y_i)^2 + \lambda (\|x_u\|^2 + \|y_i\|^2)$$

$$x_u = \left(\sum_{r_{ui} \in r_{u*}} y_i y_i^T + \lambda I_k \right)^{-1} \sum_{r_{ui} \in r_{u*}} r_{ui} y_i$$



Naive Parallel ALS

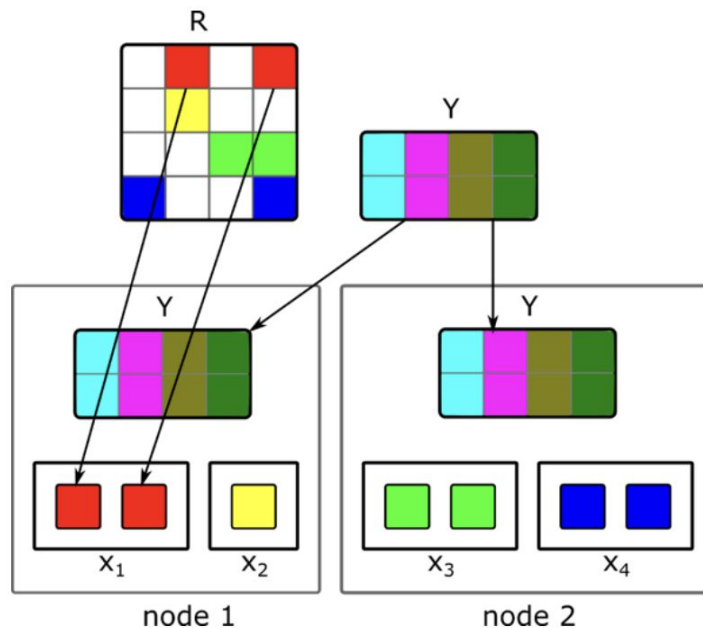
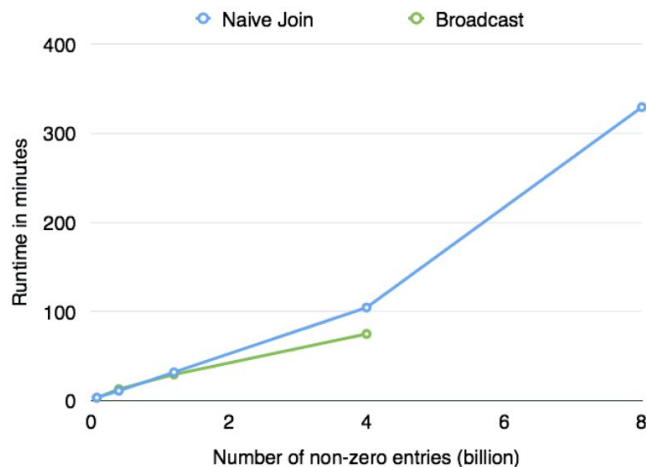
- $R = X @ Y$, update X
Join R and Y by product $\rightarrow (u, r_{ui}, y_i)$
Grouping by u
On each machine we recalculate x_u
- The same vector y_i may be needed by several users on the machine, and will be sent several times



Implementation with broadcast Y



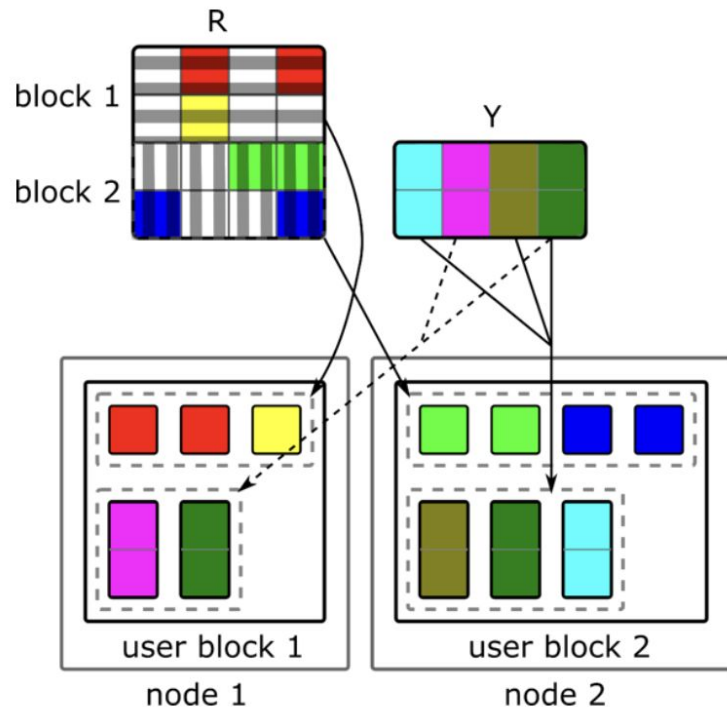
- Faster
- Has a scaling limit (Y must fit into memory)





Block ALS

- [Implemented](#) in Spark ML
- Users are fighting for blocks, each block is processed on its own machine
- We calculate once what goods will be needed in each block (mapping is stored in memory)
- Only the necessary parts of the matrices in the machine memory are sent



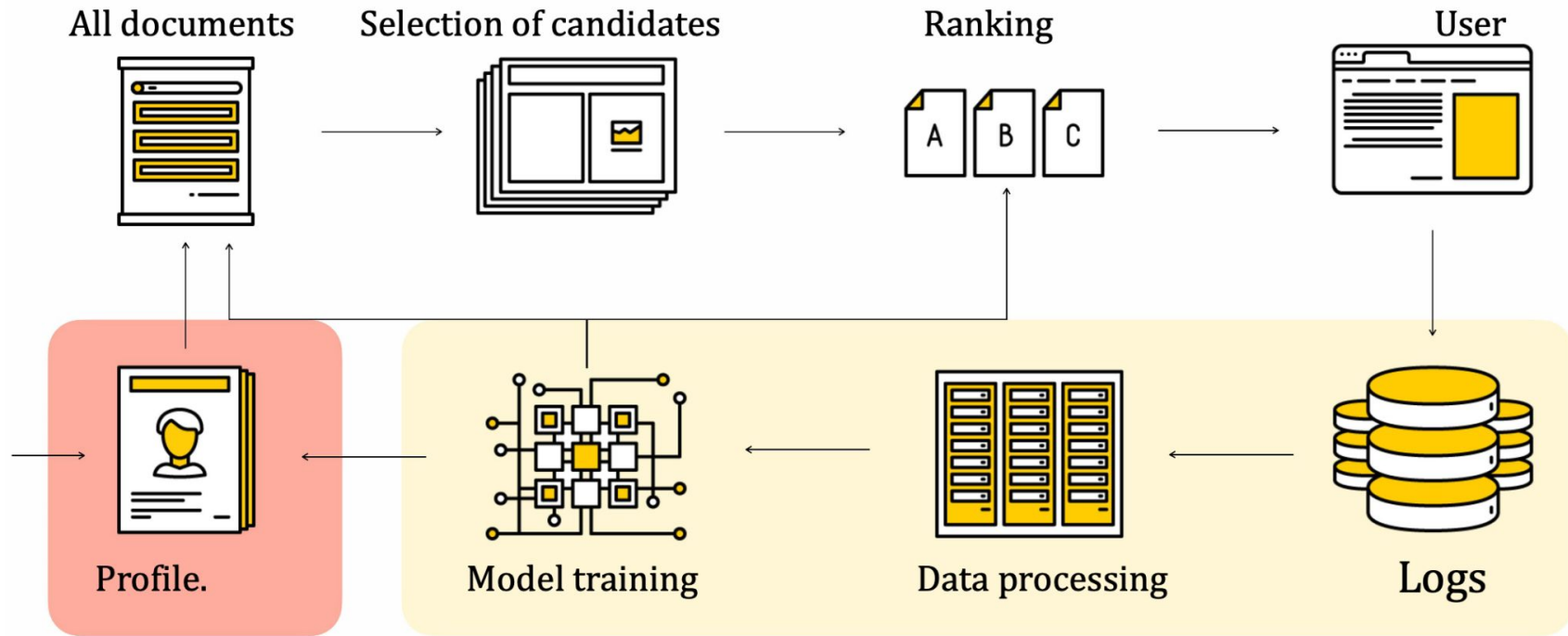
Caching during training



- Recalculate statistics (but not models) of only updating users and objects
- We save the results of third-party services that are difficult to calculate



User Profile



Fast data storage

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Fast data storage

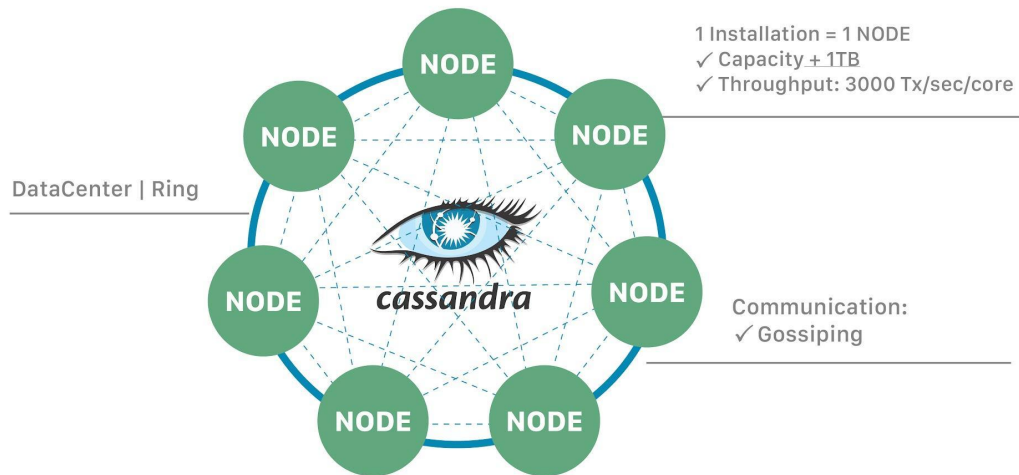
Key-value

- Reading and writing for $O(1)$
- Eventual consistency

Examples:

- Apache Cassandra
- Amazon DynamoDB
- Redis

ApacheCassandra™ = NoSQL Distributed Database



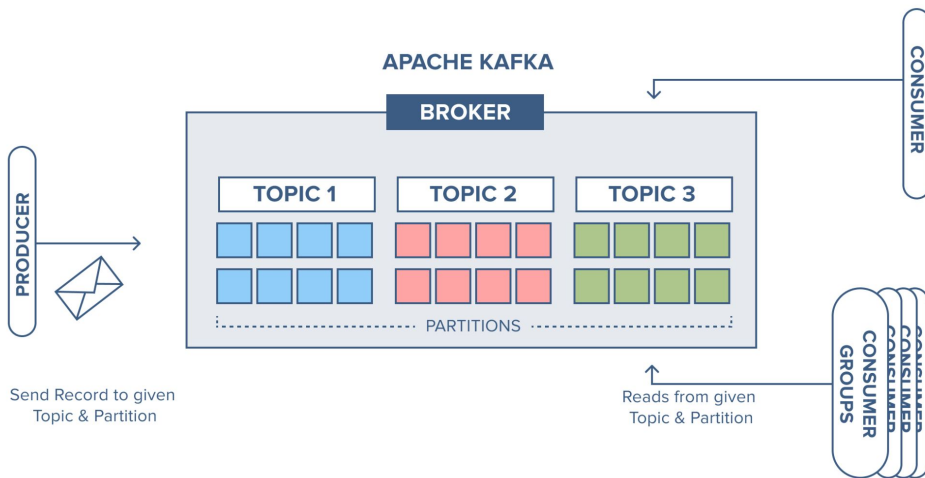


Near-realtime event handling

- We want to update user data as quickly as possible
- We update statistics, model results on the fly, reading updates from event queues

Example: writing user actions in Kafka

1. collect the user profile from the Key-Value response and events from Kafka
2. process statistics and do the ALS step every N minutes



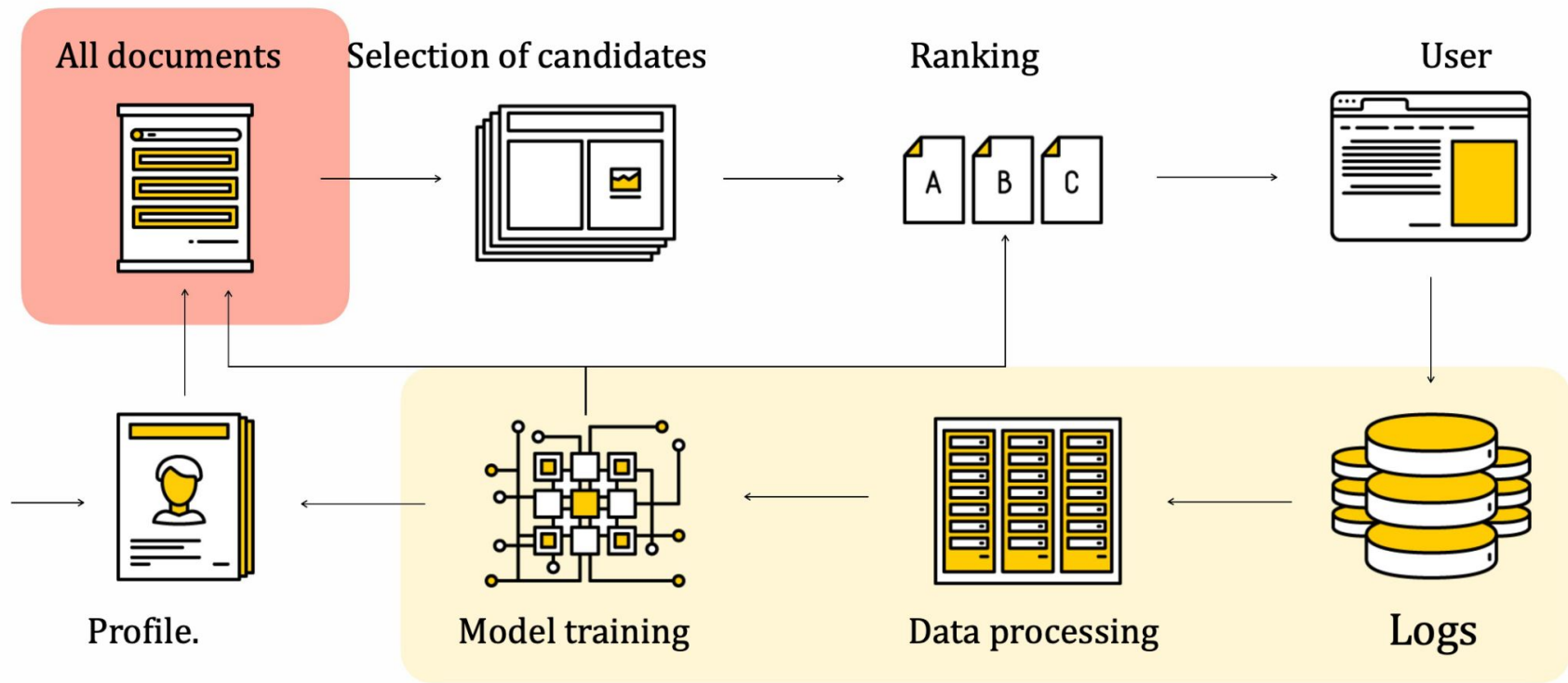
Document storage

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Document storage

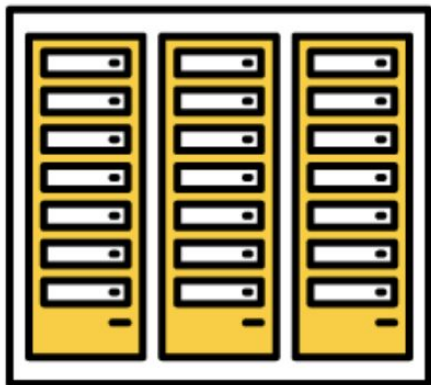


Sharding



- One server has a limit on the number of documents stored
- Let's store disjoint sets of documents on different servers, and combine the results of responses

One shard

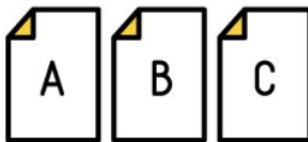
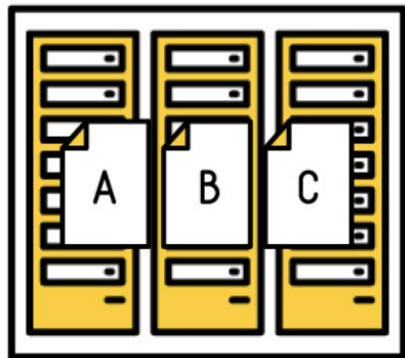


Sharding



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One shard



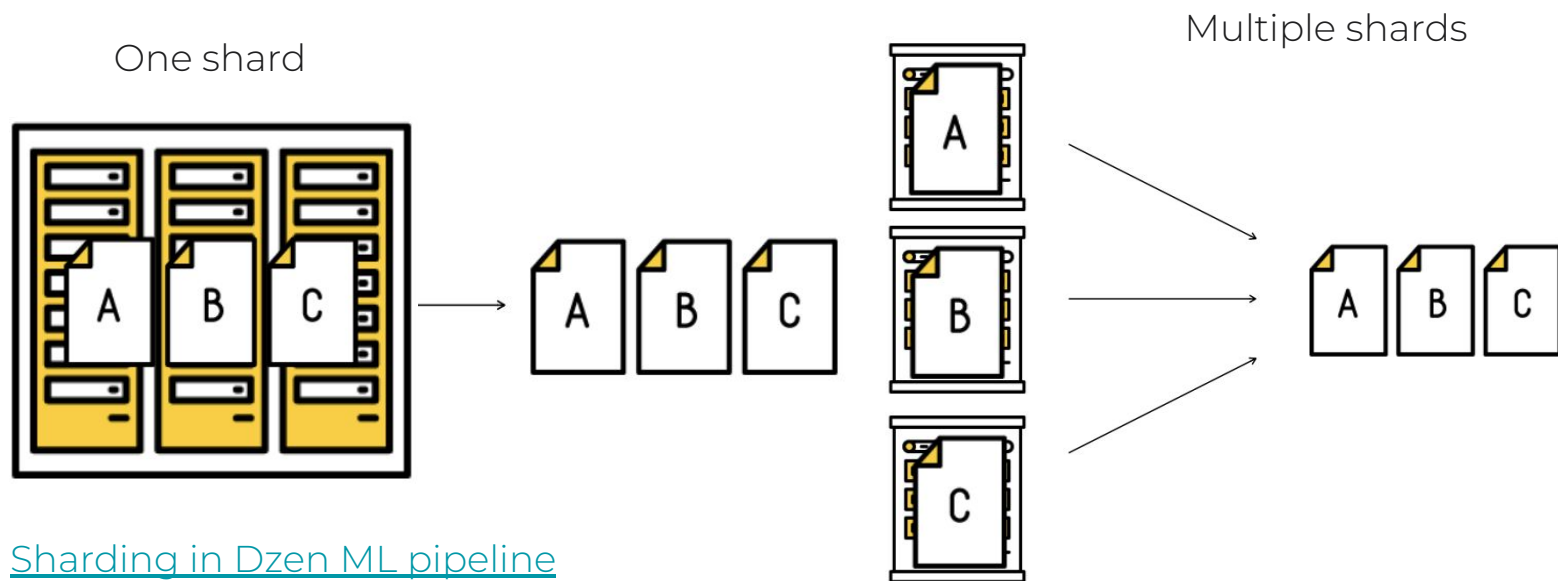
Multiple shards





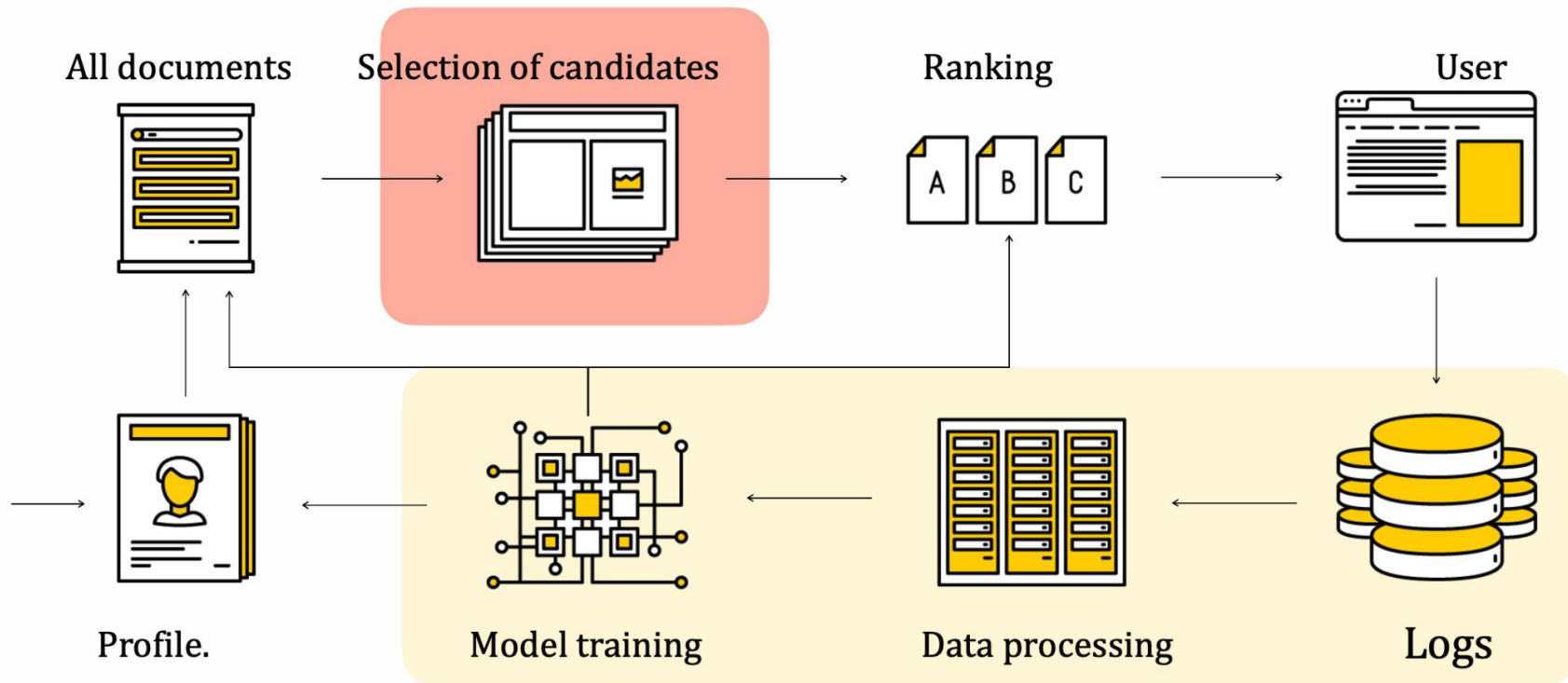
Sharding

- One server has a limit on the number of documents stored
- Let's store disjoint sets of documents on different servers, and combine the results of responses



[Sharding in Dzen ML pipeline](#)

Selection of candidates





Example: Item-based CF

- We will calculate for each product offline 1000 of the most similar to it, add it to the Key-Value storage (for example, Redis)
- Online, let's take the products evaluated by the user (the last 100) and go to Key-Value for similar ones for them

kNN indexes

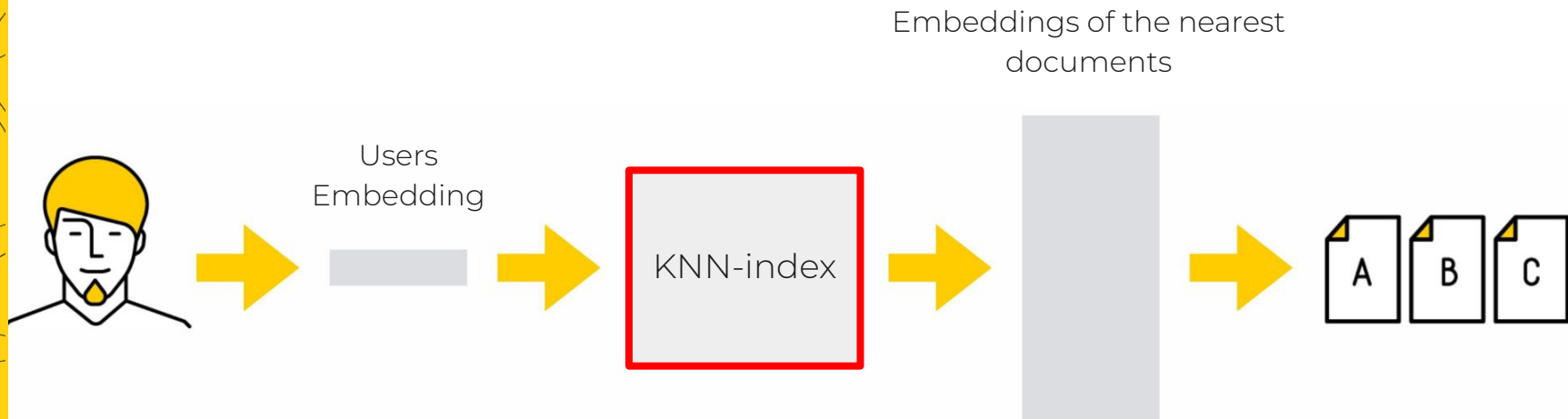
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kNN index

KNN index is a data structure that allows you to quickly find the N nearest vectors for a given vector

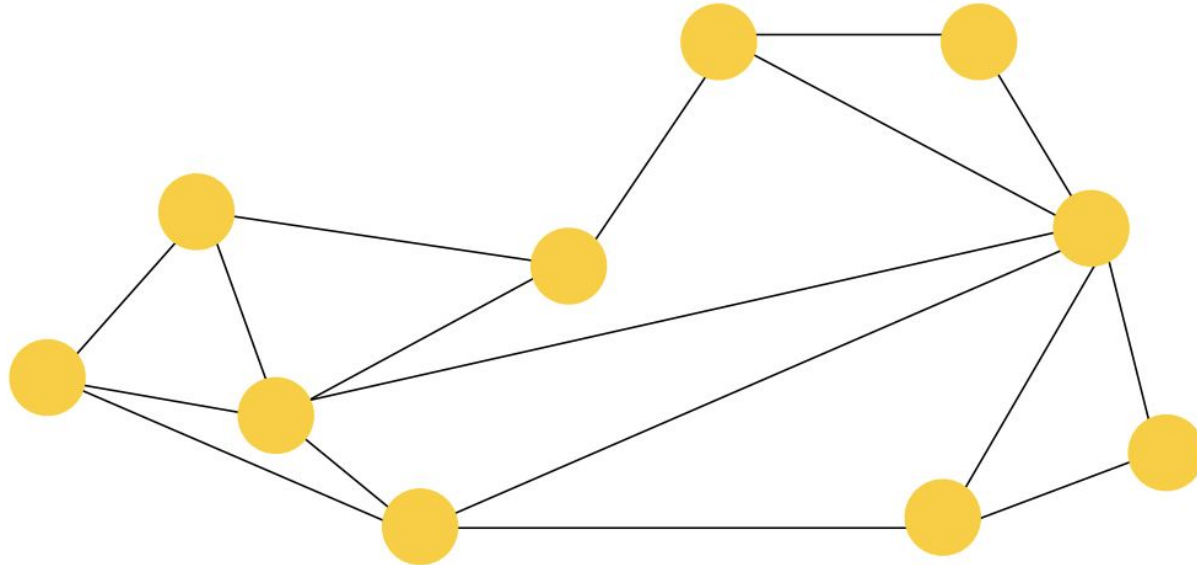


Navigable Small World

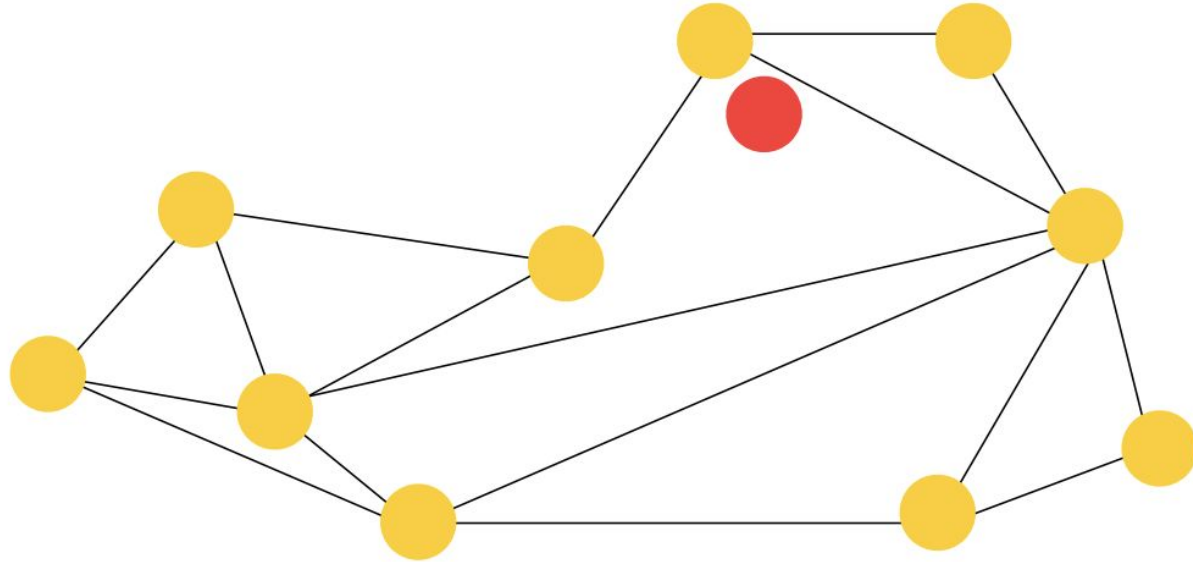


- A graph where vertices are embeddings, edges are distances
- Adding nearby and remote vertices as edges
- We start from a random vertex, count the distances and move along the graph closer to the query point

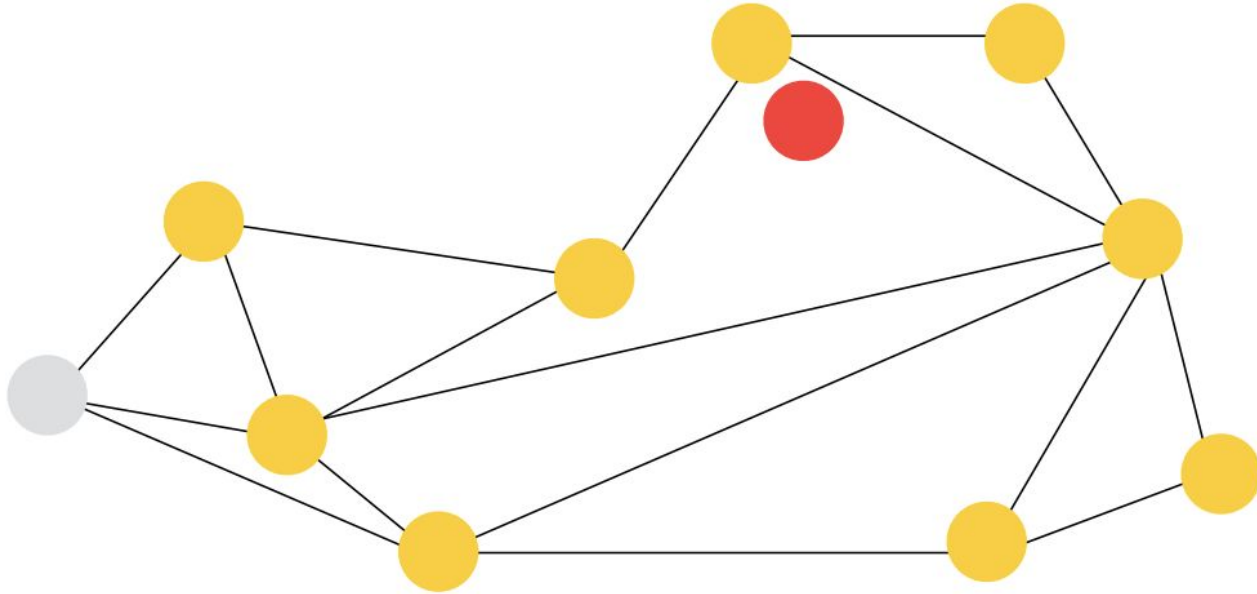
NSW (Navigable Small World)



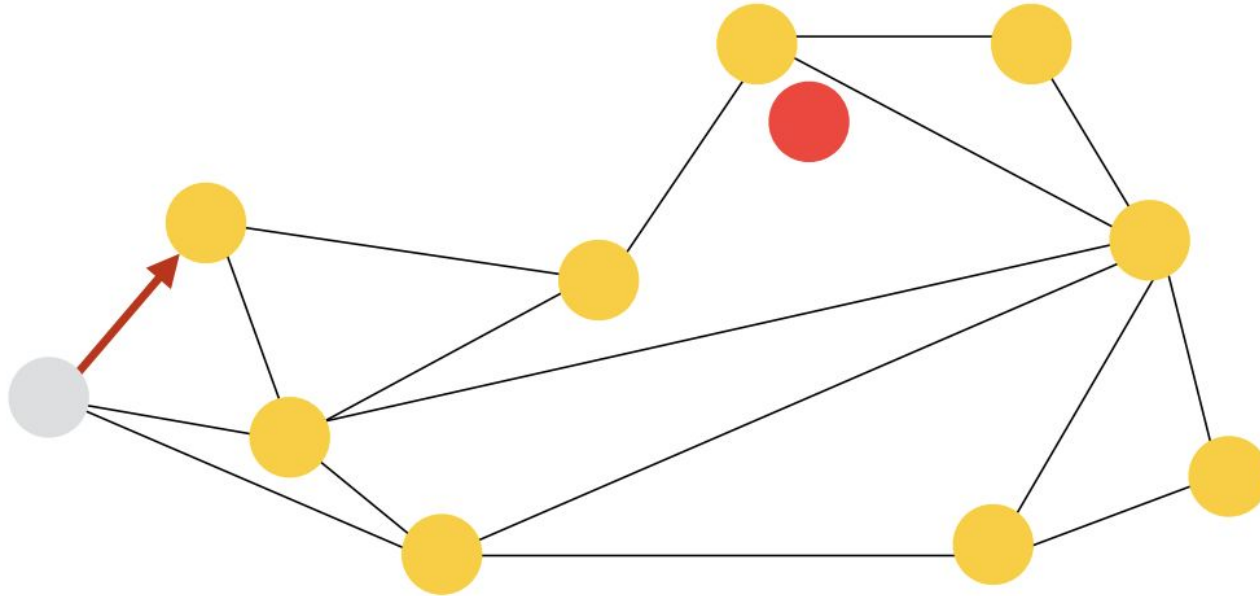
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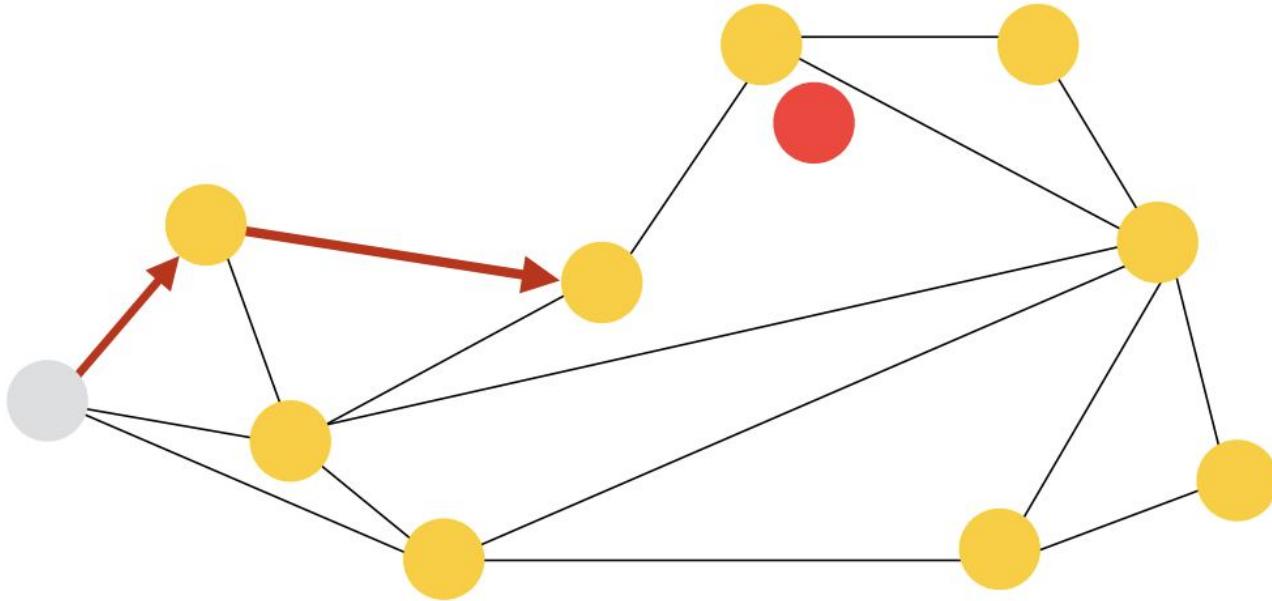
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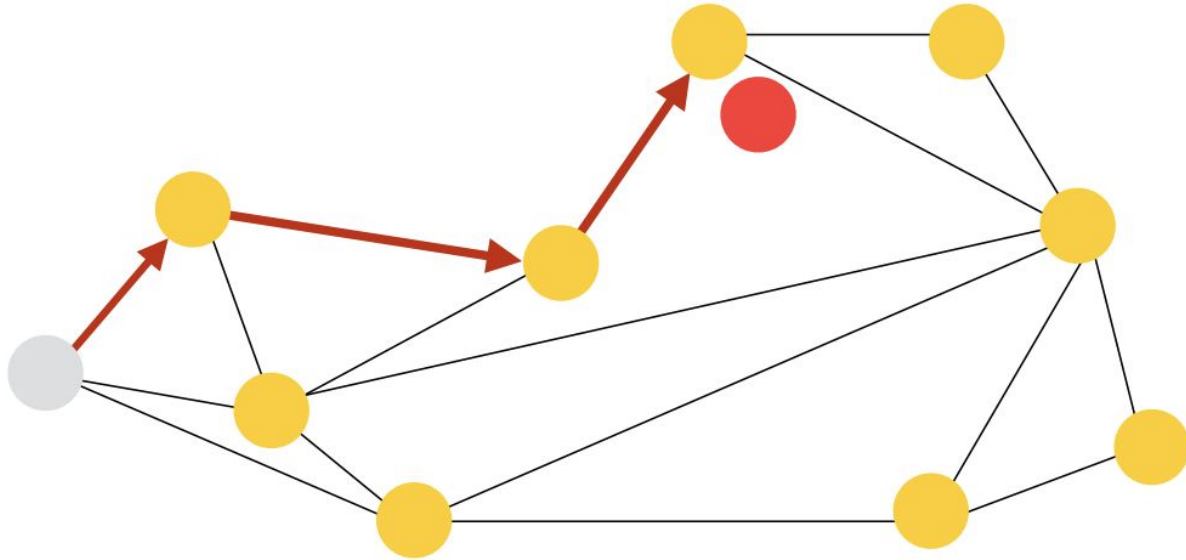
NSW (Navigable Small World)



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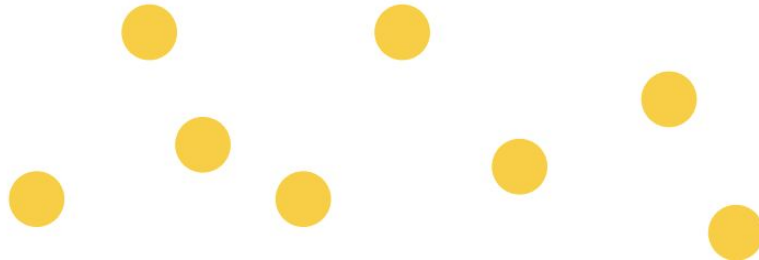


Hierarchical Navigable Small World (HNSW)

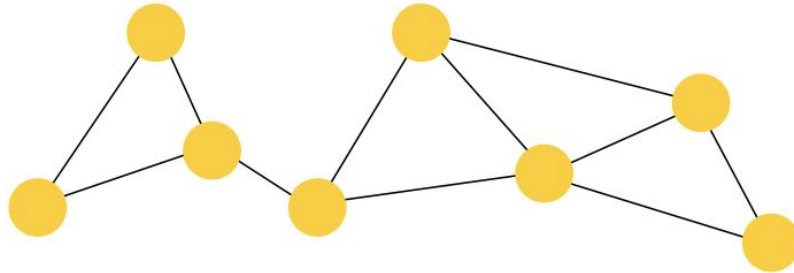
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08

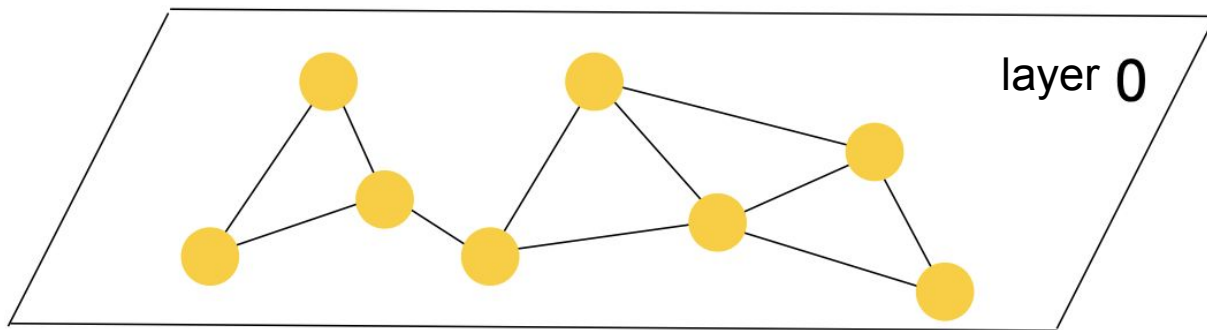
HNSW



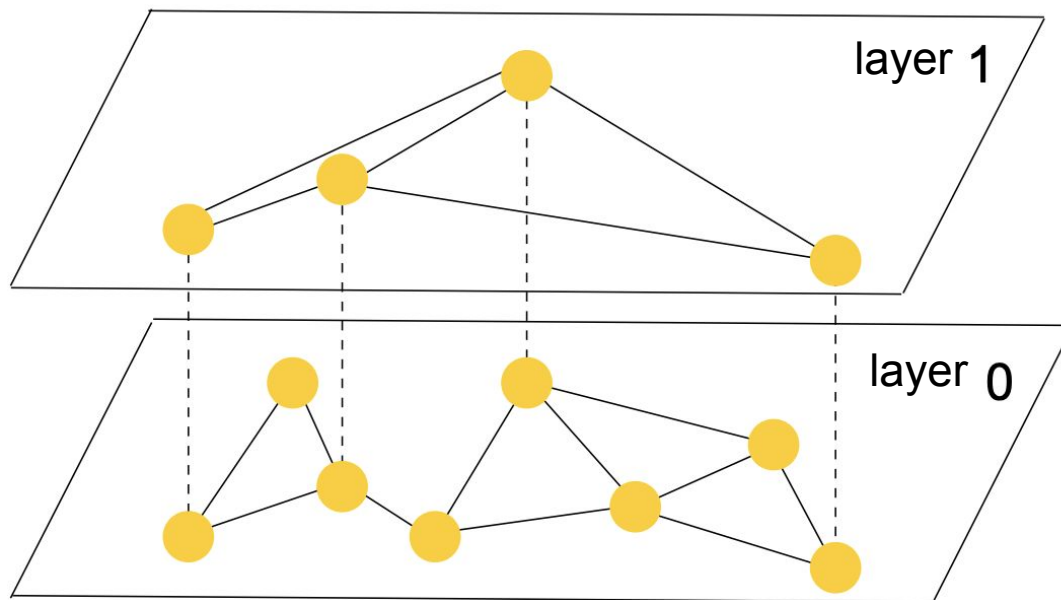
HNSW



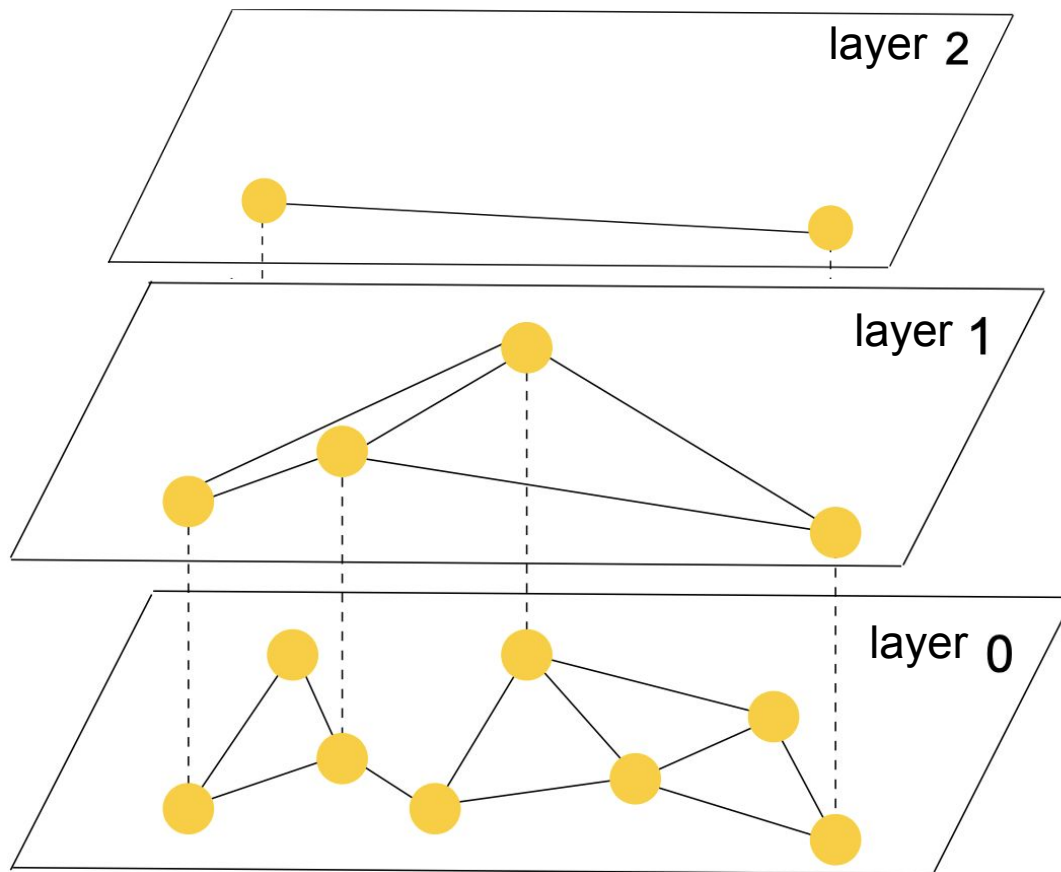
HNSW



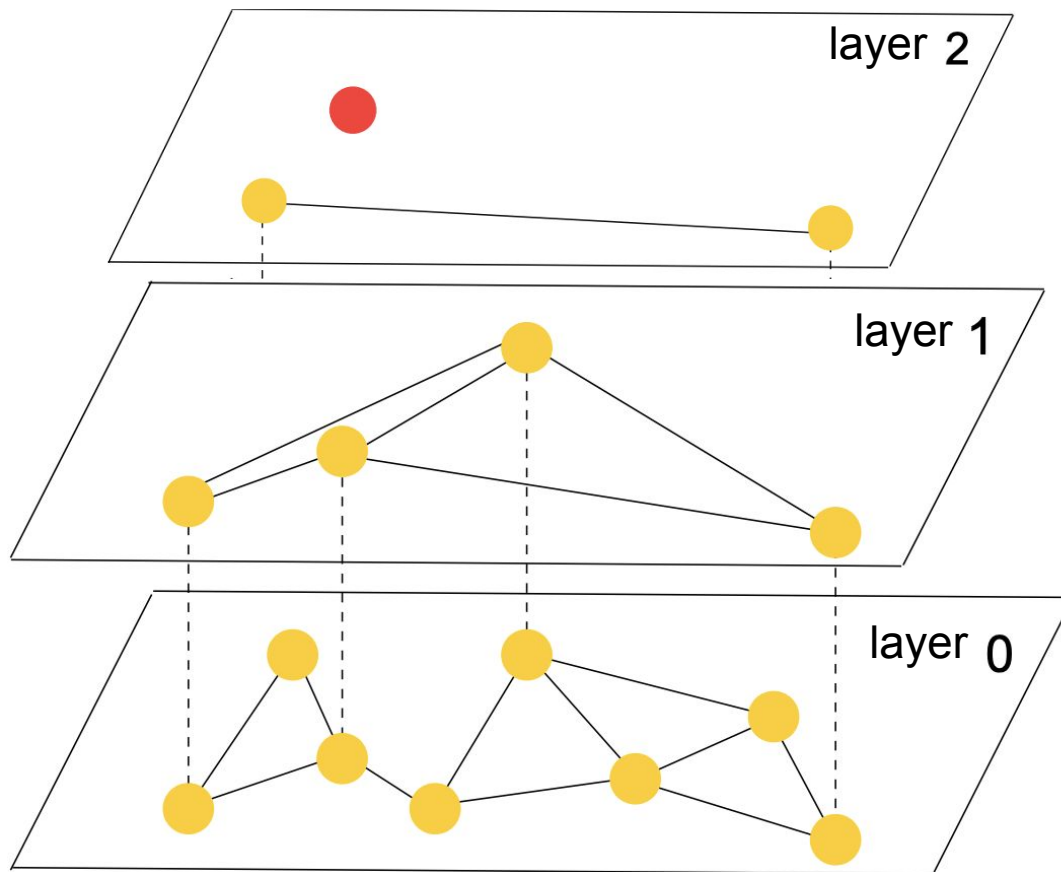
HNSW



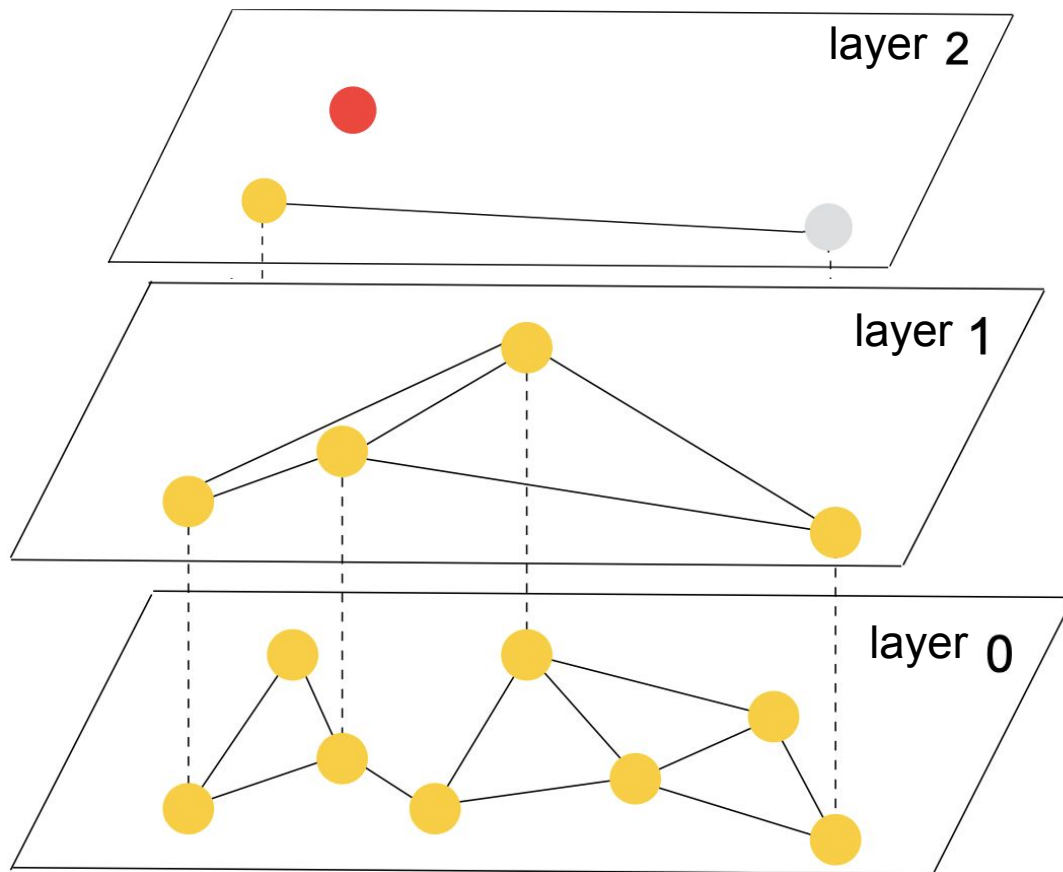
HNSW



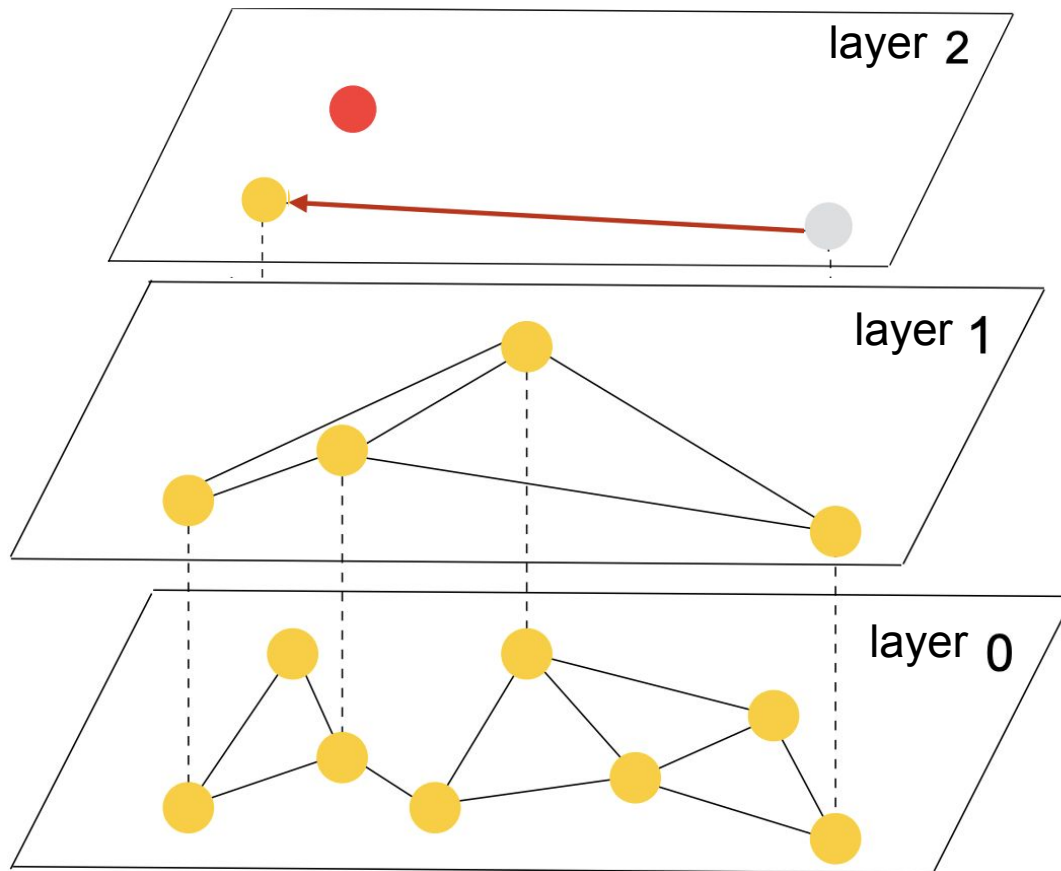
HNSW



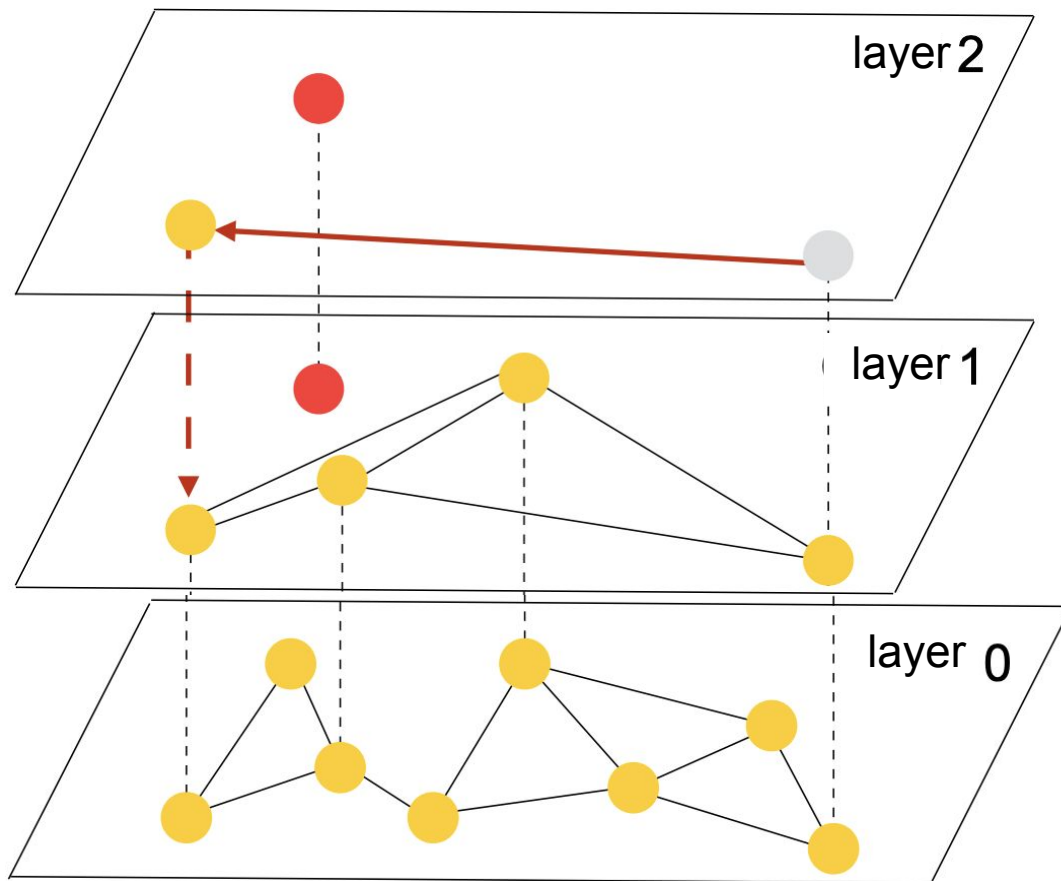
HNSW



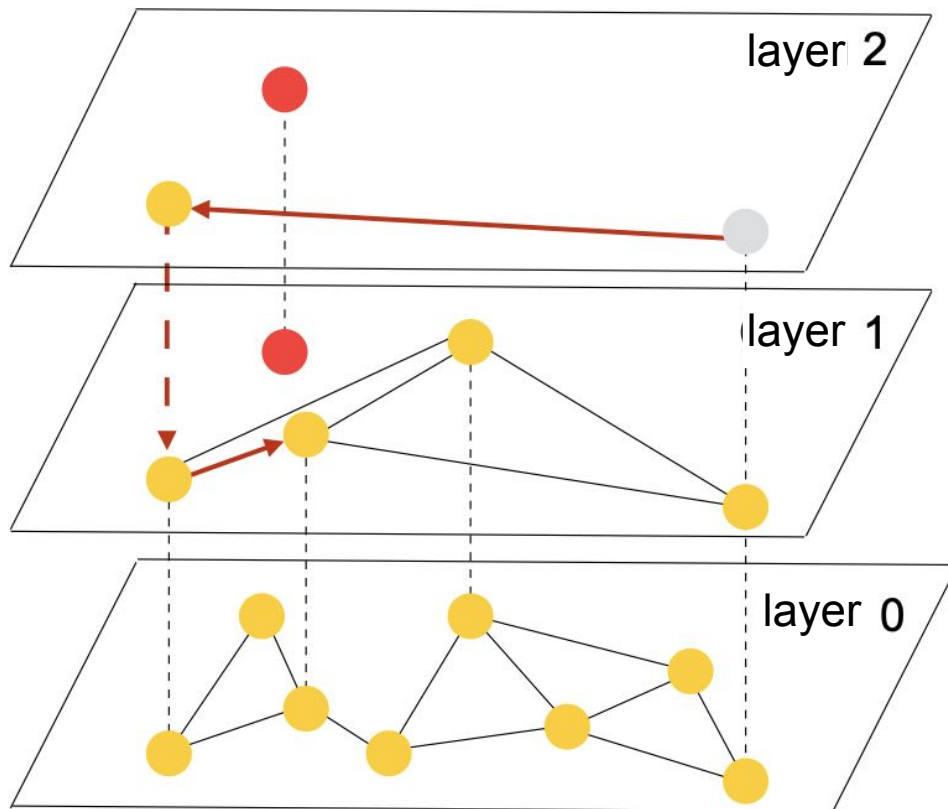
HNSW



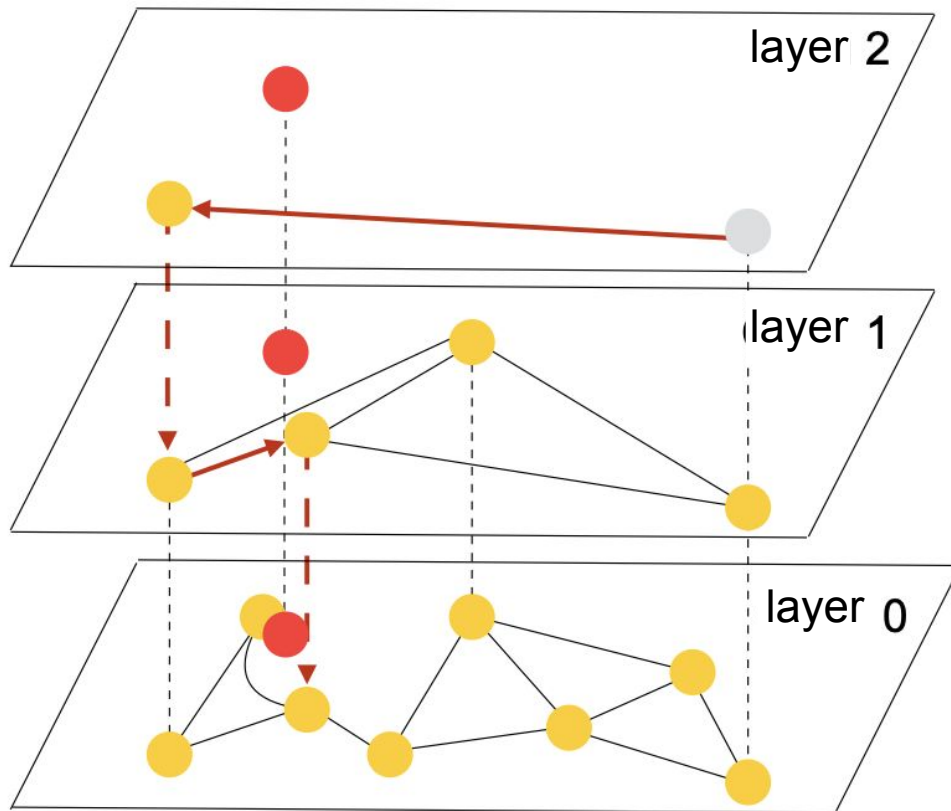
HNSW



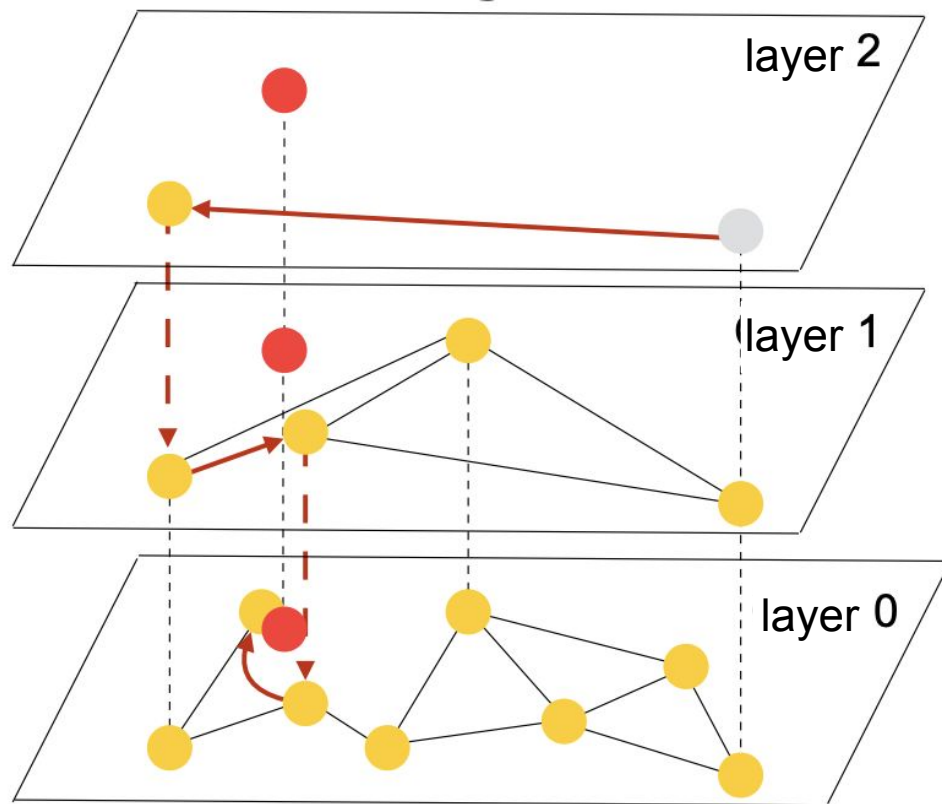
HNSW



HNSW



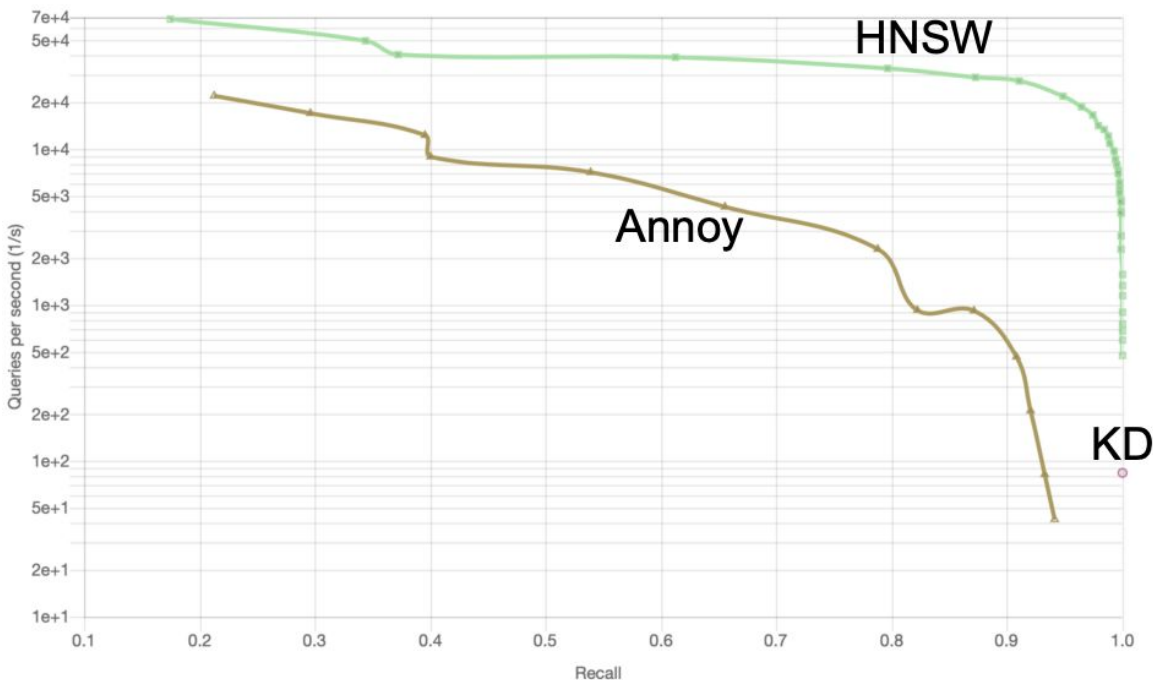
HNSW



Accuracy of approximate search



- In addition to speed, you need to find accurate nearby objects
- Comparison on ALS embedding last fm-64-dot, 360K



kNN index implementations



- <https://github.com/facebookresearch/faiss>
- <https://github.com/nmslib/nmslib>
- <https://github.com/flann-lib/flann>
- <https://github.com/spotify/annoy>
- https://lucene.apache.org/core/9_1_0/core/org/apache/lucene/util/hnsw/package-summary.html

Also list of vector databases from openAI

<https://platform.openai.com/docs/guides/embeddings/how-can-i-retrieve-k-nearest-embedding-vectors-quickly>

Ready for production solution <https://github.com/qdrant/qdrant>

Quantization

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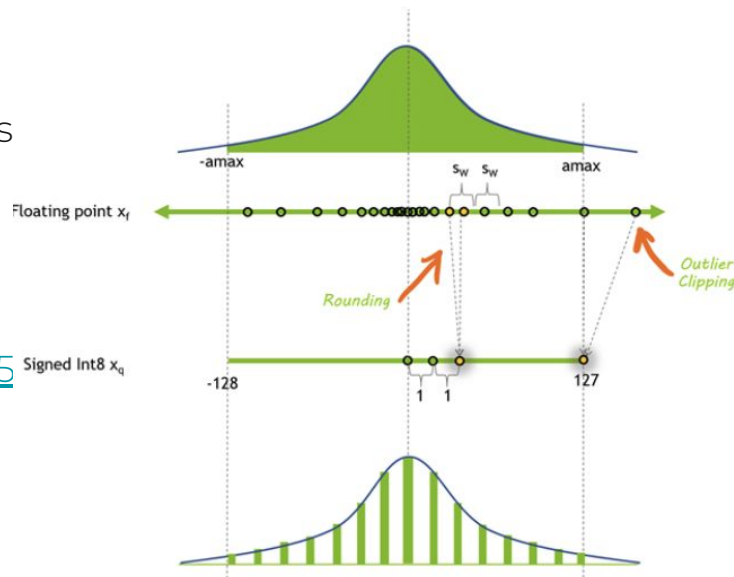
Size savings



- Embedding elements — float32
- How much will a 100-dimensional vector take in bytes?
 - 400 bites
- And if you replace it with int8?
 - 100 bites
- So let's break down the values by quantiles

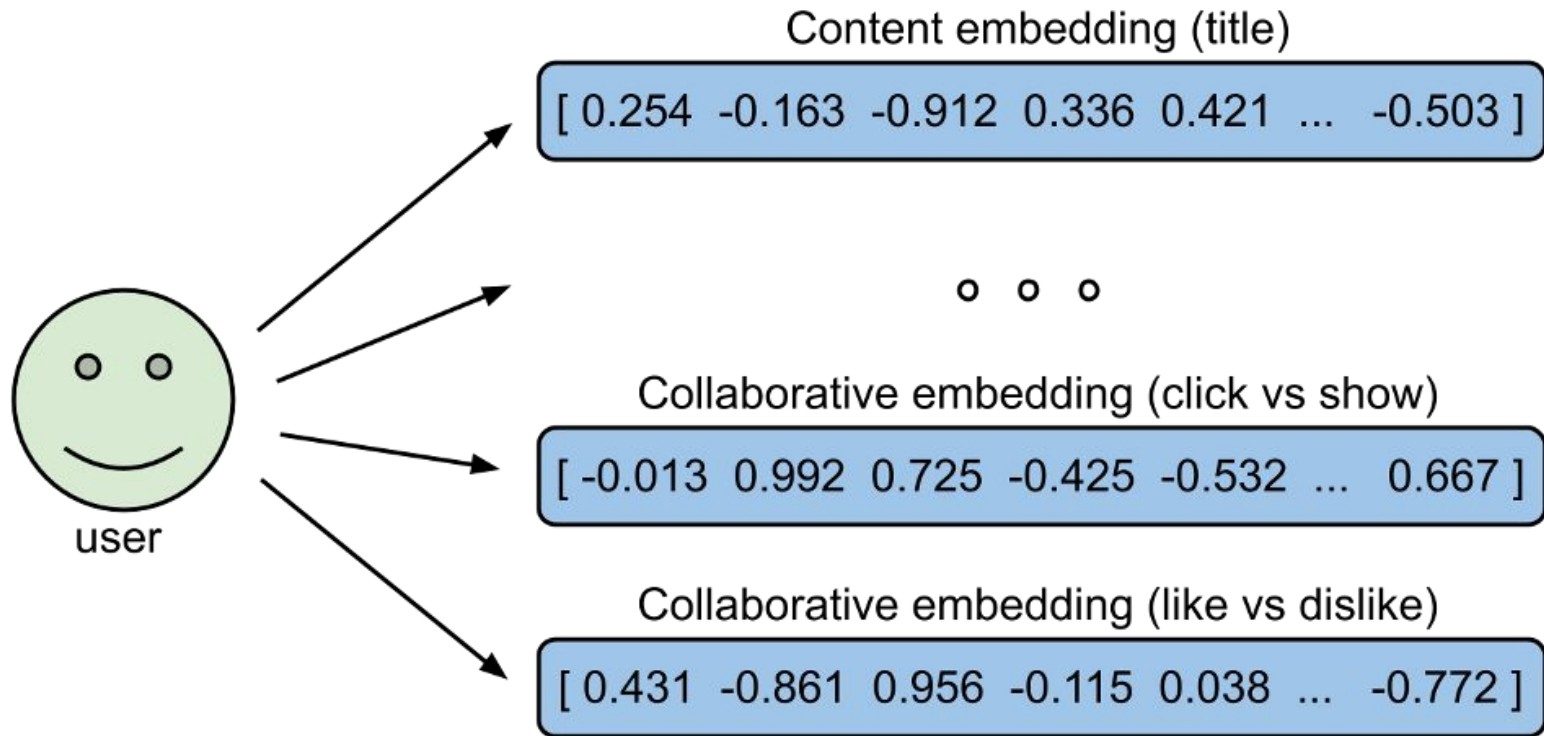
Dzen experience in quantization [ru]

<https://habr.com/ru/companies/yandex/articles/5>

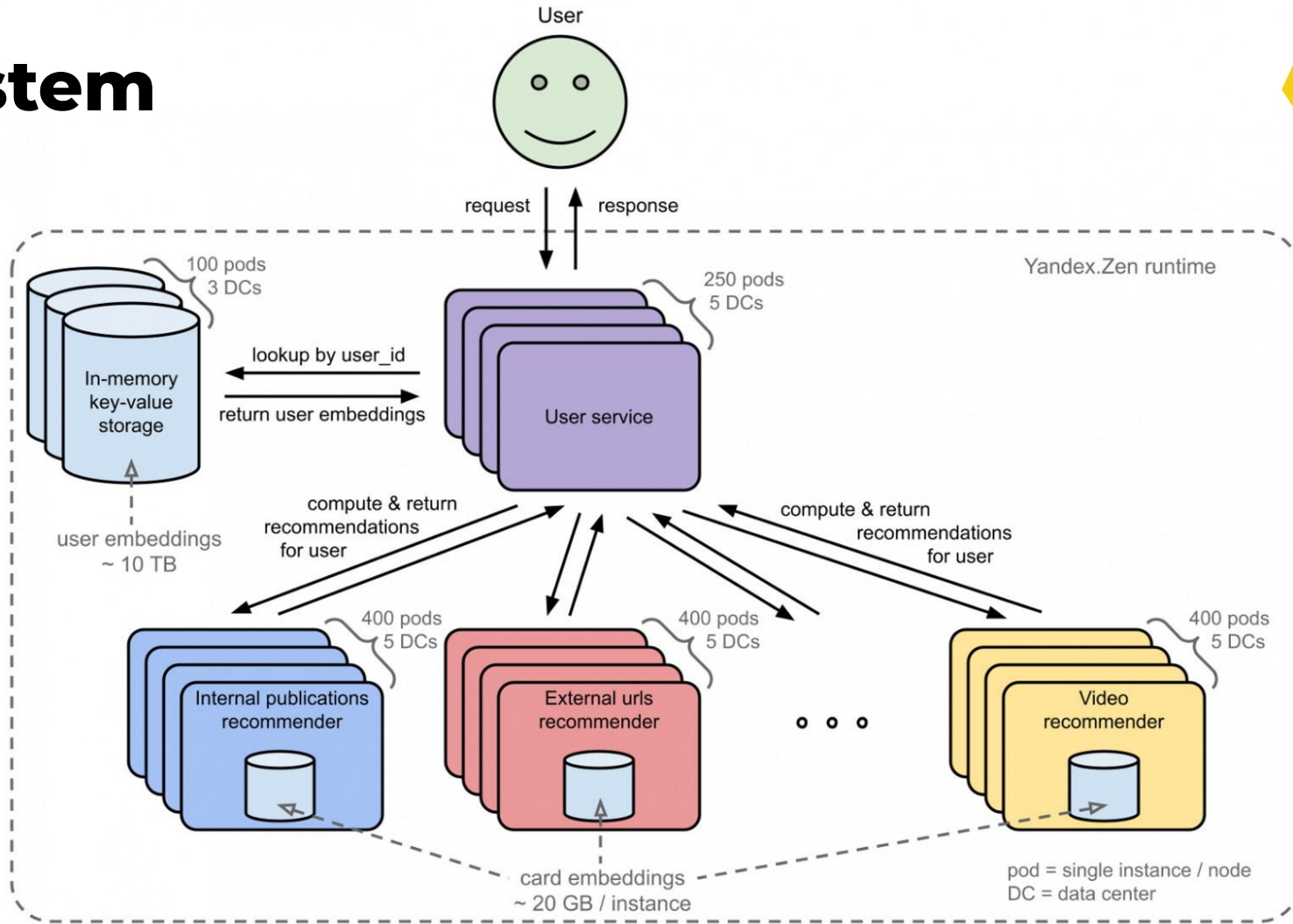




Number of embeddings

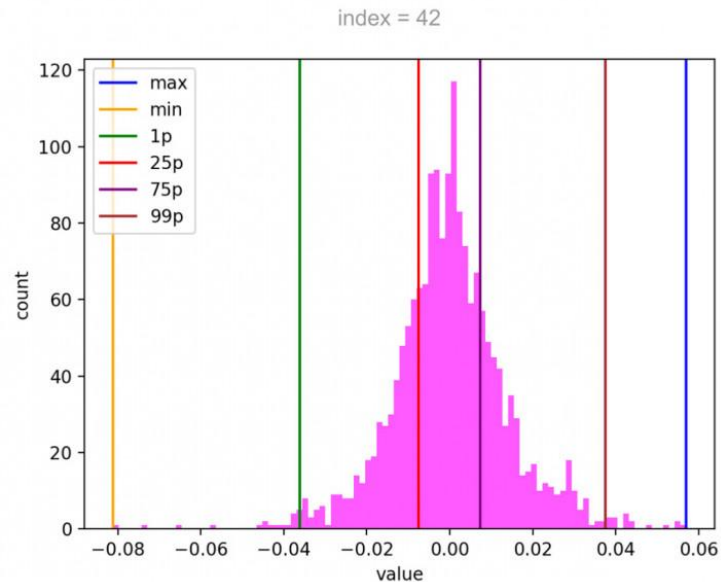
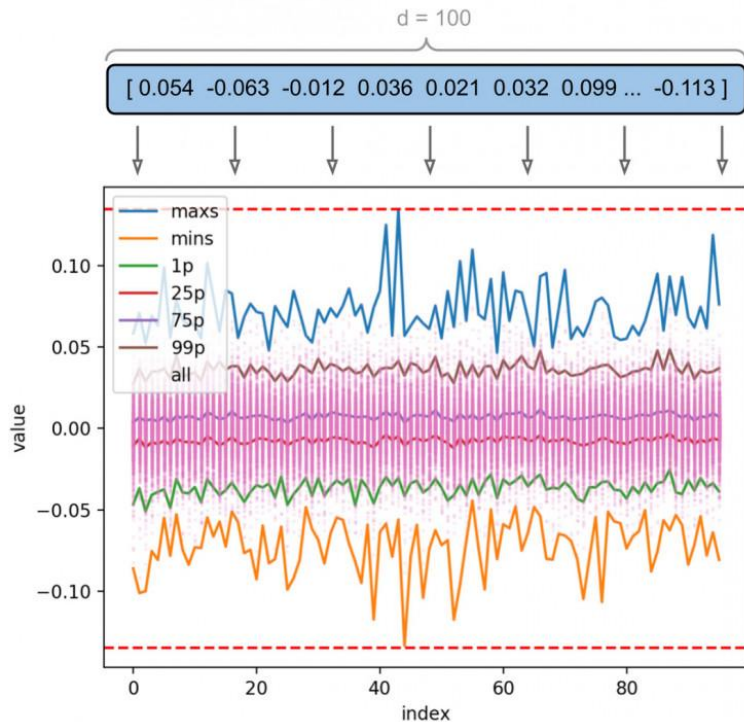


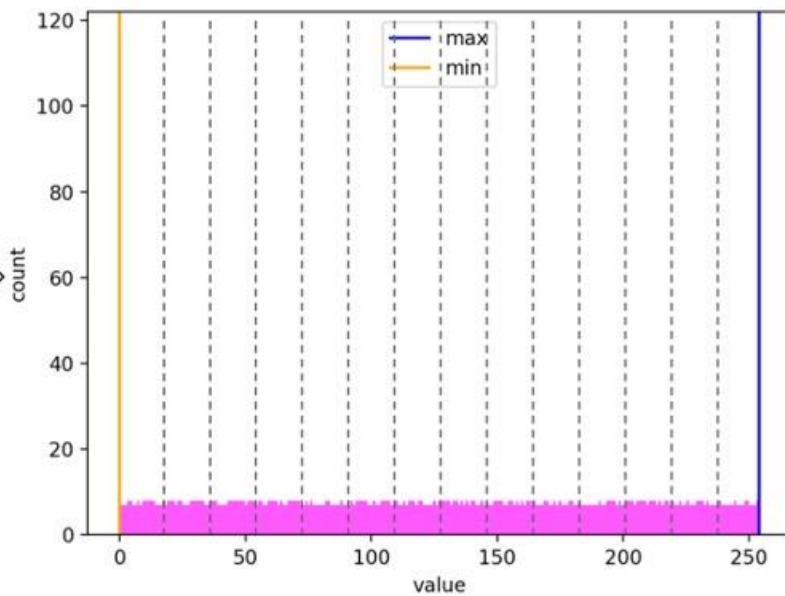
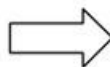
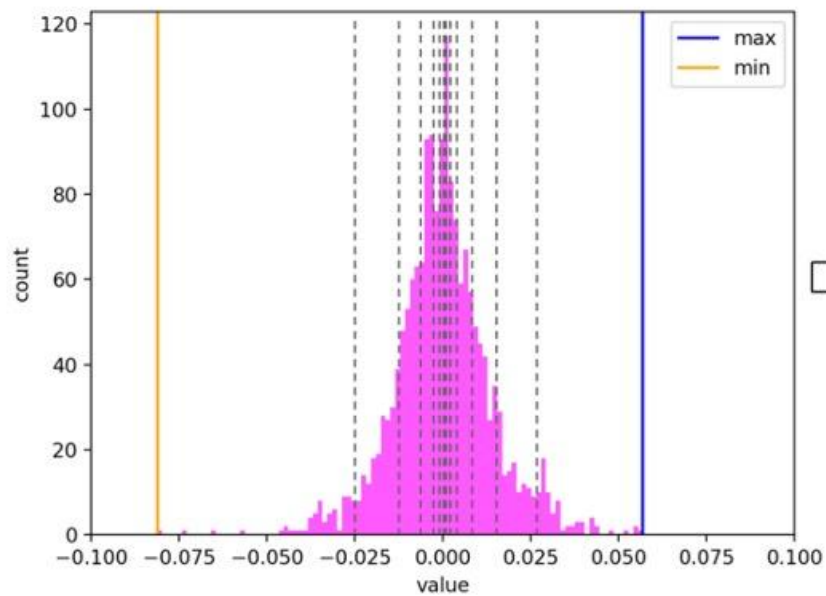
System



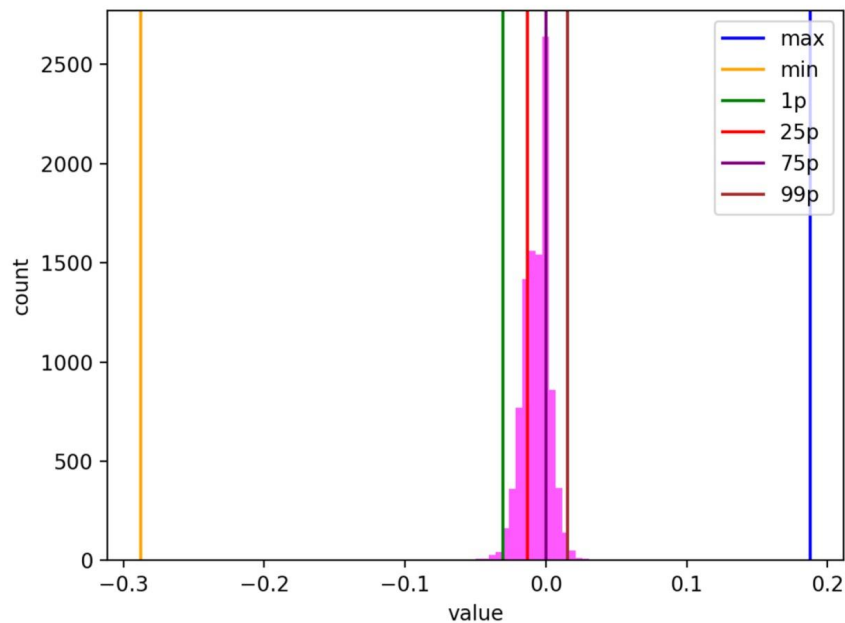


Dimension-wise distributions

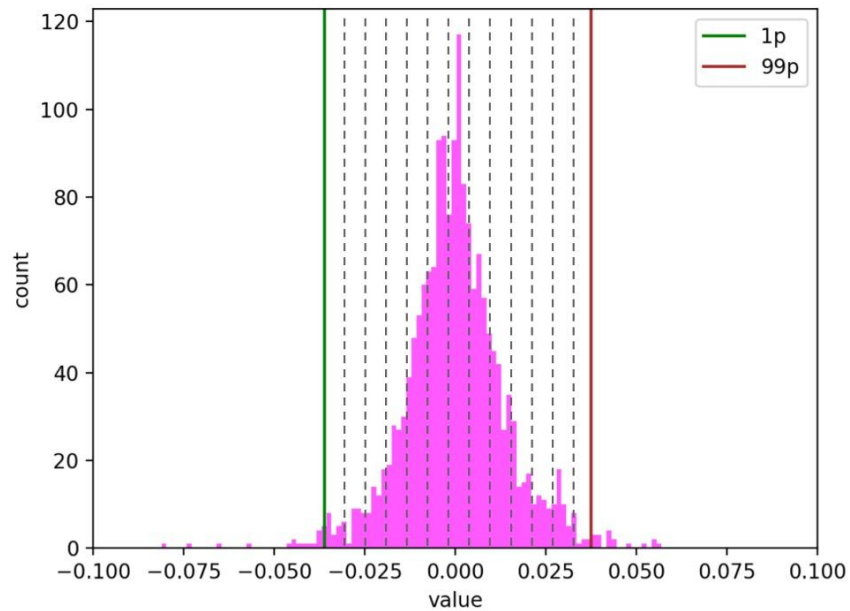




Quantization



min-max distribution



1-99 percentiles distribution

Quantization



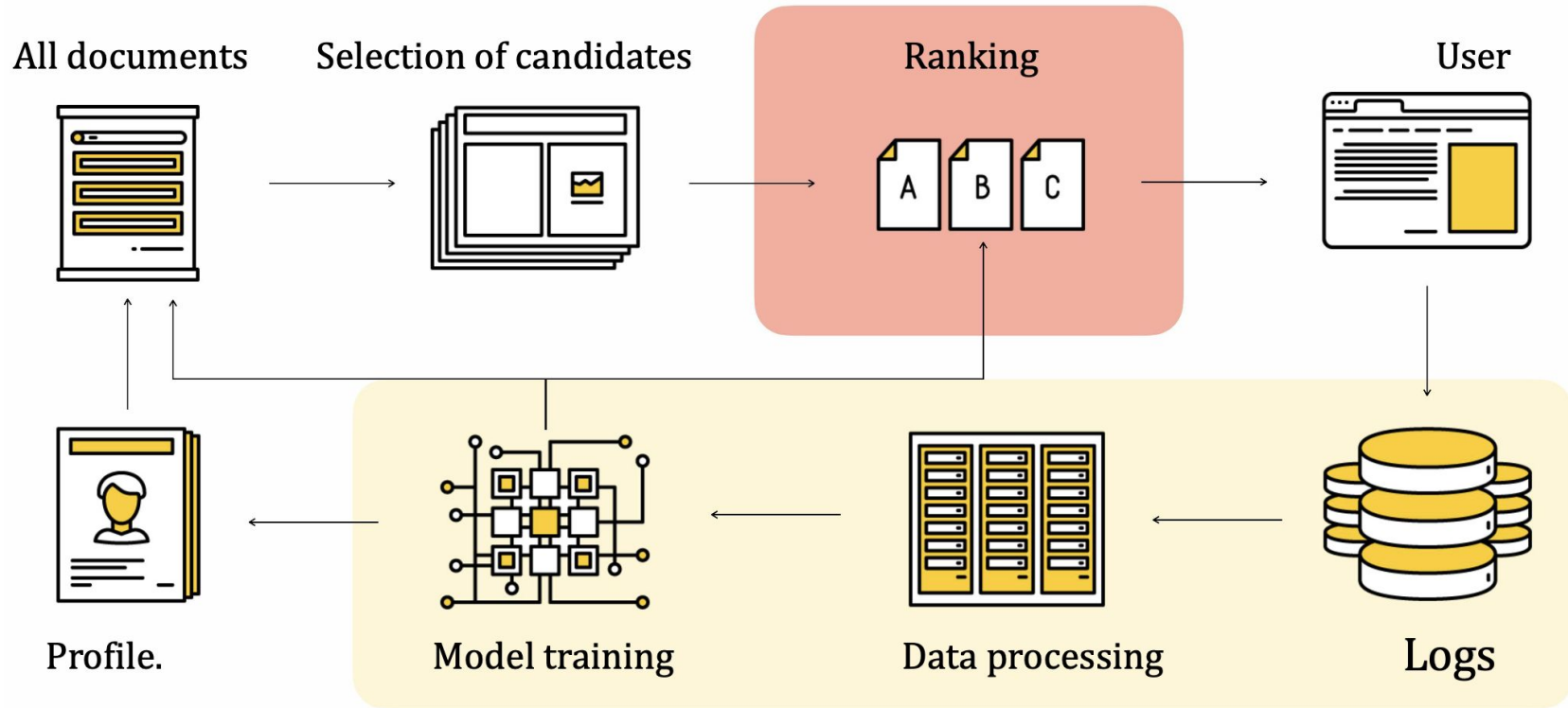
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Dzen experience in quantization (ru)

<https://habr.com/ru/companies/yandex/articles/539960/>



Ranking



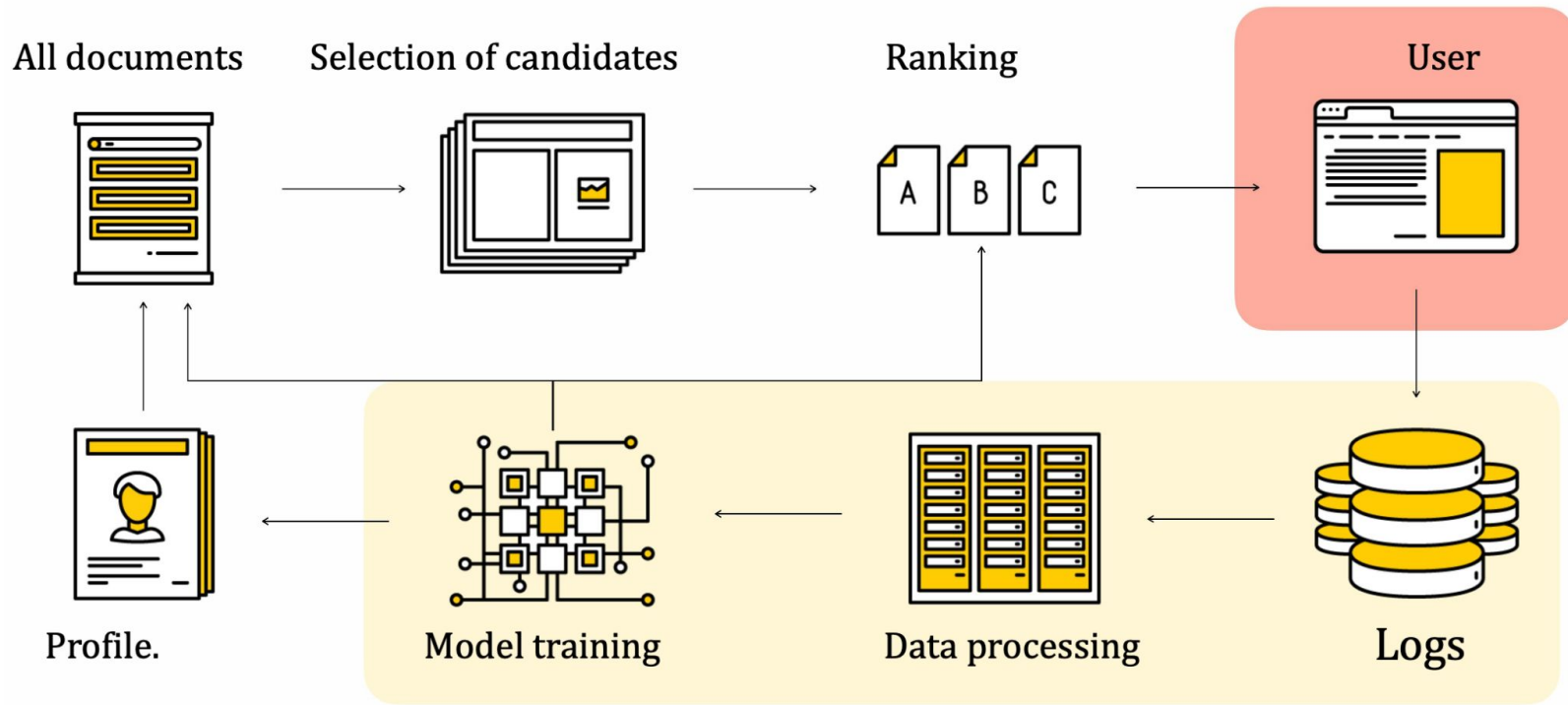
Ranking

In practice, several ranking models can be used, each of which is applied to the top of the documents selected by the previous one; gradient boosting is most often used as models. And almost always the response of the model is transformed using business rules: diversity, history of interaction with the document, freshness, time of day, etc.





Load from users



Caching during serving



- We get the results from the cache, reduce the load on the main service

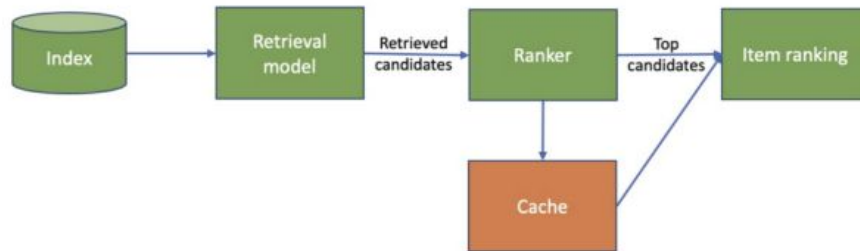


Figure 1: A Delivery Flow of Recommendation Systems with Simple Caching Strategy

- Adding a model that predicts cache usage

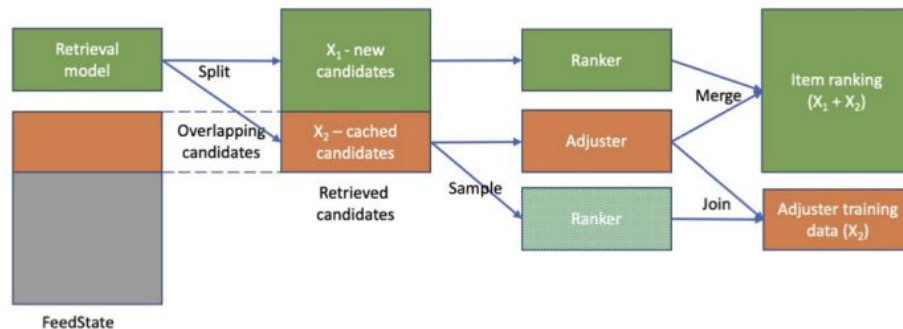


Figure 2: A Delivery Flow with Smart Cache

Recommendation system lifecycle

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Features logging



- It is necessary to train the model on the data recorded during the operation of the real system
- Collecting data on historical casts is guaranteed to lead to a discrepancy in values when training and applying models
- Errors in calculations will give rise to incorrect signs with leaks of the target variable ("face")



Reproducible learning

Always versioning your experiments

- Data
 - Parameters of models and training
 - Executable code
-
- And of course fix Random Seed (I advise 42)

Username: **vkokhtev**

Date: 2021-10-21 10:22:35

Status: Completed

Tags: **general** **general_ranking**

Model Name: CatBoost-YetiRank

Model Options:  {

"border-count" : 128

"depth" : 8

"iterations" : 1000

"l2-leaf-reg" : 0.0167177222323797

"learning-rate" : 0.2599701978545636

}

Dataset Name: **CENSORED**

Experiment: 735fdf4b-bf92-44b1-b79b-e2d6cfbd9a86



Metrics monitoring

- Metrics from the last lecture must be constantly monitored
- Model quality metrics

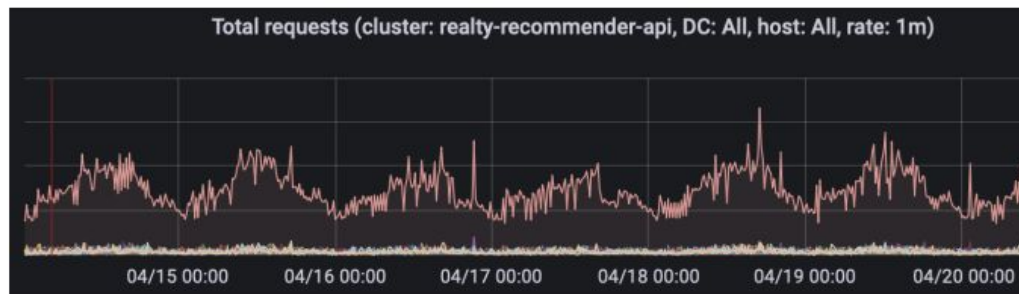
Technical

- Number of requests, response time
- Values of features and responses of models

Количество пользователей, смотревших сниппеты



Total requests (cluster: realty-recommender-api, DC: All, host: All, rate: 1m)



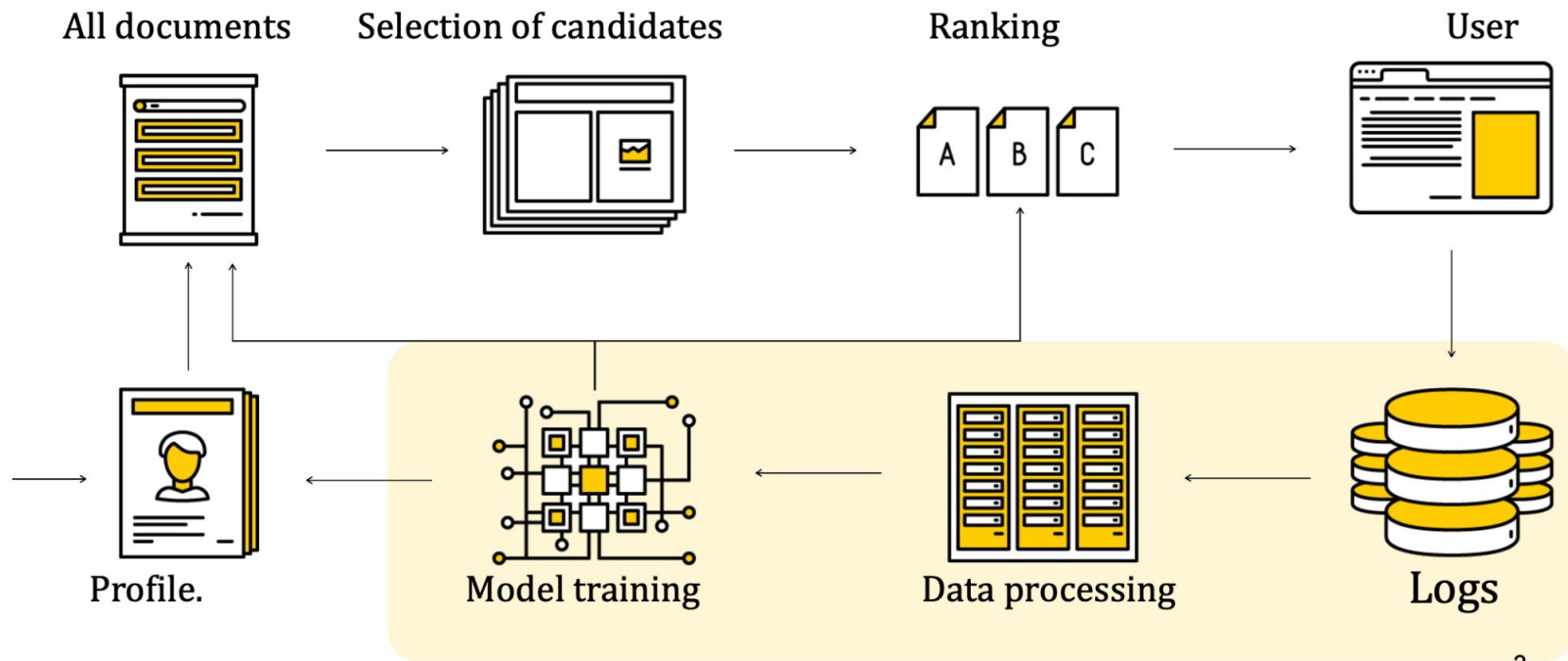
Regular models retraining



- Even if you have received a good recommendation model once, it will become outdated over time. Therefore, it is necessary to periodically train/retrain models on fresh data
- You can trust automation and run each regular model in an A/B experiment, and if the main metrics are stable or improved, include them for all users



Resume



Thanks for attention!

Questions?

girafe
ai

